Scientific and Information Visualization of Summer-2019 US Meteorology (gridMET) and a Century of Global Natural-Disaster Records (EM-DAT)

DAS732 Data Visualization – Programming Assignment 2 (A2) · Team Pythagoras

Lakshya Jain BT2024044 · Kalpit BT2024093 · Arnav BT2024233

International Institute of Information Technology Bangalore, Bengaluru, India

***Abstract*—This report presents the six visualization tasks of A2. For SciVis we use the gridMET daily meteorological dataset for our allotted slice, July–September 2019, sampled uniformly every ten days (ten time instances). Daily maximum temperature and precipitation are colour-mapped with sequential, rainbow, diverging, discrete and logarithmic schemes under global and local normalisation; vapour pressure deficit is contoured with our own vectorised Marching Squares implementation, which matches the contourpy reference to 3×10^{-14}°, and with filled contours; and the 10 m wind is shown with quiver plots and streamlines under different grid-sampling and seeding strategies, complemented by 4 km close-ups of three storms that fall between the samples. Animated GIFs accompany every task. For InfoVis we use our A1 dataset, the EM-DAT natural-disaster country profiles, from which we derive (i) a 167-node network of countries with similar hazard regimes, compared under five NetworkX layouts, explored in an interactive viewer and exported to Gephi, (ii) a multivariate table shown in a JavaScript parallel-coordinates plot with cluster-based edge bundling, brushing and axis reordering, and four hierarchies shown in a zoomable treemap with six tiling algorithms, and (iii) yearly, five-year and decadal series shown as horizon and ridgeline plots, including a share anomaly that separates changes in the hazard mix from the growth of reporting. The visualizations expose both physical signals—a heat wave, Tropical Storm Imelda, an early-season cold outbreak, the seasonal retreat of dry air—and data artefacts such as quantisation ties, reporting bias in disaster damage and state succession.**

***Index Terms*—colour mapping, marching squares, quiver plot, streamlines, node-link diagram, graph layout, parallel coordinates, edge bundling, treemap, horizon chart, ridgeline plot, gridMET, EM-DAT.**

I. Introduction

A2 asks for three SciVis tasks on a scalar/vector meteorological field and three InfoVis tasks on the dataset used in A1. Every member owns exactly one SciVis and one InfoVis task (Table I); data download, cleaning and processing were done jointly, as permitted. Section II describes both datasets, the choice of period and time instances, and all preprocessing. Sections III–V cover colour mapping, contour mapping and vector-field visualization; Sections VI–VIII cover the node-link diagram, the parallel coordinates and treemap, and the horizon and ridgeline plots. Section IX discusses what the tasks have in common and their limitations. Every figure is stored at full resolution in the `images/` folder under its figure index (e.g. `Fig16_….png`), together with the GIF animations; the code and a README are in `code/`.

TABLE I
Task ownership

Member	*SciVis task*	*InfoVis task*
Lakshya Jain (BT2024044)	Colour mapping (Sec. III, Figs. 2–9)	Parallel coordinates with edge bundling and zoomable treemap in JavaScript (Sec. VII, Figs. 28–36)
Kalpit (BT2024093)	Contour mapping (Sec. IV, Figs. 10–15)	Node-link diagram and layout comparison with NetworkX (Sec. VI, Figs. 23–27)
Arnav (BT2024233)	Quiver plots and streamlines (Sec. V, Figs. 16–22)	Horizon and ridgeline plots (Sec. VIII, Figs. 37–43)

II. Datasets and Data Handling

A. gridMET slice and the choice of period

gridMET [1], [2] provides daily surface meteorology for the contiguous United States (CONUS) on a 1/24° (≈4 km) latitude–longitude grid of 585 × 1386 cells. Our team was allotted **July–September 2019** (92 days). The quarter is well suited to the three SciVis tasks because it spans the transition from the peak of summer to early autumn: the area-weighted CONUS mean of the daily maximum temperature falls from about 30 °C in July to 22 °C at the end of September, the mean vapour pressure deficit drops by about 40 % and the mean 10 m wind speed rises by about 60 % as the first autumn troughs arrive (Fig. 1). The window also contains the peak of the Atlantic hurricane season and several well-documented extremes [8]: Hurricane Barry (Louisiana, 13–15 Jul), the central and eastern heat wave of 18–21 Jul, Hurricane Dorian along the Carolinas coast (5–6 Sep), Tropical Storm Imelda (south-east Texas, 17–19 Sep) and an unusually early snowstorm and cold outbreak in the northern Rockies (27–30 Sep). The slice therefore offers both a seasonal trend and sharp synoptic features.

Five variables were used (Table II): daily maximum temperature *tmmx* and precipitation *pr* for colour mapping, daily mean vapour pressure deficit *vpd* for contour mapping, and, as required, wind speed *vs* and direction *th* at 10 m for the vector-field task. The data were obtained from the gridMET THREDDS NetCDF Subset Service, which serves the same values as the yearly `<var>_2019.nc` files listed under “Direct download of NetCDF files”, but lets us request only what is needed: (i) the ten sampled days at full resolution (16 MB per variable); (ii) all 92 days of temperature, VPD and wind at every fourth cell (≈16 km), whose domain means and percentiles a 16 km subsample estimates well; (iii) all 92 days of precipitation at full

resolution, because storm maxima are only a few cells wide and the 16 km subsample underestimates the daily maximum by a median of 12 % and up to 50 % (Imelda: 528 instead of 707 mm on 18 Sep); and (iv) wind and precipitation at full resolution over the Gulf and south-east coasts on the storm days between the samples (Sec. V-G). The server returns uncompressed NetCDF-3, so products (iii) and (iv) are stored with zlib compression. All fields are processed with xarray [5] and plotted with matplotlib [6].

TABLE II
GRIDMET VARIABLES AND SAMPLED TIME INSTANCES

Variable	*Units (storage step)*	*Used in*
tmmx – daily max. temperature	K → °C (0.1)	Colour mapping
pr – daily precipitation	mm/day (0.1)	Colour mapping (log/discrete)
vpd – mean vapour pressure deficit	kPa (0.01)	Contour mapping
vs, th – 10 m wind speed, direction	m/s (0.1), ° from N (1)	Quiver, streamlines
Sampled days t1–t10 (2019): 01, 11, 21, 31 Jul; 10, 20, 30 Aug; 09, 19, 29 Sep (every 10th day from 1 Jul)		

B. Choice of time instances

We sampled **uniformly, every 10th day from 1 July**, which gives the maximum permitted ten instances and covers 90 of the 92 days. Uniform sampling was preferred to random or event-targeted sampling for four reasons: (i) a constant interval means that the perceived rate of change in an animation is proportional to the real rate of change, which random gaps would distort; (ii) ten days is longer than the 3–7-day lifetime of a weather system, so consecutive frames are nearly independent and together trace the seasonal march rather than a single storm; (iii) uniform samples are unbiased for quarter-level statistics, which we need for the global colour range, the anomaly baseline and the contour areas; and (iv) the choice is reproducible and was made before looking at the maps.

Fig. 1 places the samples within the daily record and shows what the sampling captures and what it misses. The heat wave is captured by t3 (21 Jul): the mean maximum temperature over the north-east was 32.9 °C, one day after its 33.2 °C peak. Tropical Storm Imelda is captured by t9 (19 Sep, 295 mm/day at 30.2° N, 93.8° W), although its 707 mm/day peak (29.9° N, 94.2° W) fell on 18 Sep. The northern-Rockies cold outbreak is captured by t10 (29 Sep, the coldest day of the quarter in Montana, with a mean maximum of 3.2 °C). The samples miss Hurricane Barry's rainfall maximum (365 mm on 14 Jul in south-western Louisiana), Hurricane Dorian's passage along the Carolinas coast (251 mm on 5 Sep) and a 413 mm Florida Big Bend event on 17 Aug. These are temporal aliasing of short-lived extremes, an unavoidable cost of ten-day sampling; Sec. V-G therefore shows the three storms in close-ups. All maxima are taken from the full 4 km grid.

C. Preprocessing of the gridMET fields

- **Masking.** 40.3 % of the grid (ocean, Canada, Mexico) is missing. NaNs are kept as masks throughout. Statistics use cos φ area weights; areas use per-cell areas (CONUS land = 7.95 M km²). Marching Squares skips cells with a missing corner.
- **Units and quantisation.** Temperature is converted from K to °C. VPD is stored in 0.01 kPa steps and precipitation in 0.1 mm steps. Both matter: exact ties at isovalues (Sec. IV) and empty histogram bins below 0.3 mm (Fig. 8d).
- **Day convention.** A gridMET day ends at midnight Mountain Standard Time (07 UTC the next day), so an evening storm in the East can fall on the following day's map.
- **Projection.** All maps use a Lambert conformal conic projection (standard parallels 33° and 45° N, central meridian 96° W). Because it is conformal, arrow directions are preserved, but only if the vectors are handed over correctly. cartopy [7] interprets vectors given in longitude–latitude coordinates as rates of change of longitude and latitude, and a degree of longitude is only cos φ as long as a degree of latitude. We therefore divide the eastward component by cos φ and rescale the pair to the true speed before cartopy rotates it into the projection. Without this step diagonal winds are turned towards the meridian, by a median of 5.0° and up to 12.0° on t10 (11.7° for a north-easterly at 49° N); with it, the projected directions agree with a finite-difference reference to better than 0.01° and speeds are unchanged. All quiver plots and streamlines use the corrected vectors. The map extent is fitted to the valid cells.
- **Anomalies** are taken relative to the per-cell mean of the ten samples, which under uniform sampling is an unbiased estimate of the quarter mean.
- **Wind vectors** follow the meteorological convention: *th* is the direction the wind blows *from*, clockwise from north, so

$$u = -v_s \sin\theta, \quad v = -v_s \cos\theta. \tag{1}$$

For streamlines the field is interpolated onto a regular grid in projected metres, and grid points more than 18 km from any valid land cell are masked so that no streamline is drawn over water.

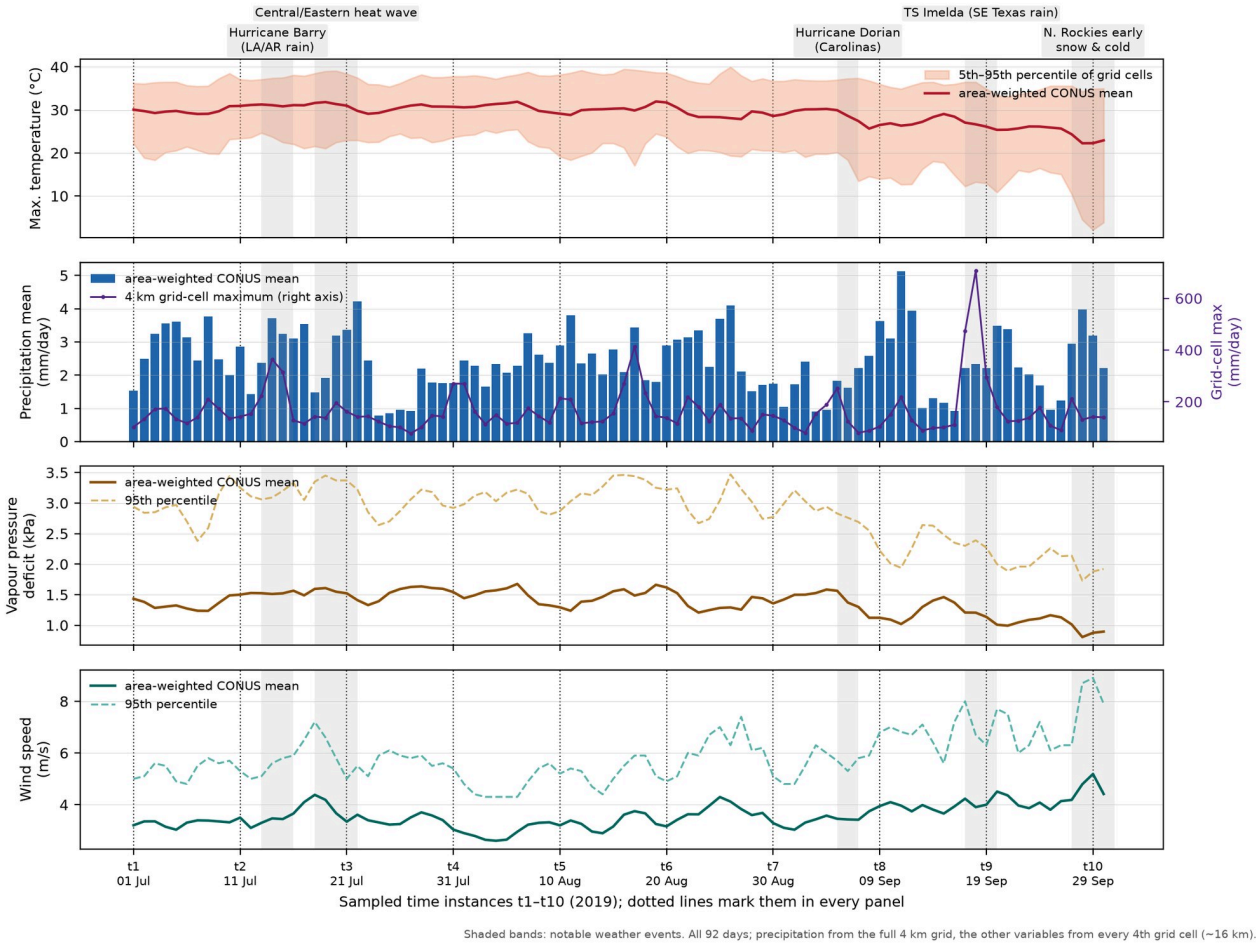


Fig. 1. Daily CONUS statistics for 1 Jul–30 Sep 2019 (all 92 days; precipitation on the full 4 km grid, the other variables on every 4th grid cell) with the ten uniformly sampled days t1–t10 (dotted lines) and five notable weather events (grey bands). Maximum temperature: area-weighted mean and 5th–95th percentile band; precipitation: area-weighted mean (bars) and grid-cell maximum; vapour pressure deficit and wind speed: mean and 95th percentile.

D. EM-DAT country profiles (A1 dataset)

The A1 dataset is the EM-DAT [3] country-profile export on Kaggle [4]: 10,431 rows × 13 columns, 225 entities, 1900–2023, one row per year × country × disaster type × subtype. Each row gives the number of events, people affected, deaths, damage in original and CPI-adjusted US$, and the CPI. All rows belong to the "Natural" group (5 subgroups, 13 types, 25 subtypes). EM-DAT records an event if it caused at least 10 deaths, affected at least 100 people, or led to an emergency declaration or a call for international assistance [3]. Table III lists the data problems we found and how each was handled.

TABLE III
DATA-QUALITY ISSUES IN THE EM-DAT COUNTRY PROFILES AND THEIR HANDLING (NUMBERS FROM EMDAT_CLEANING_LOG.TXT)

Issue	*Evidence*	*Handling*
Format	";" separator, decimal comma in CPI; header typo "Disaster Subroup"; trailing spaces ("Extreme temperature ")	explicit parser options; columns renamed; labels stripped
Missing subtype	2,133 rows (20.4 %), mostly floods and storms	kept as "<type> (unspecified)"; for the network, unspecified floods (1,928 events) and storms (694) re-assigned in proportion to the country's specified subtypes
Missing impacts	affected 27.3 %, deaths 29.3 %, damage 63.2 % of rows blank	treated as "not reported" (NaN), never as zero; sums use only reported values; the share of records with a damage figure becomes an analysis variable (Sec. VII)
Damage adjustment	adjusted = original × 100 / CPI (CPI = 100 in 2022); 2023 rows have no CPI	relation verified (max. relative error 2.0×10^{-4}); 4 values for 2023 kept at base-year prices
Partial year	2023 has 75 events vs. 388 per year in 2018–2022 (snapshot of 6 Apr 2023)	2023 excluded from all temporal analyses and from the treemap
Sparse early record	only 9.3 % of events occurred before 1970	cross-sectional analyses use 1970–2022; long-run views keep 1900–2022 and are interpreted with care
Historical entities	9 entities (USSR, Yugoslavia, Czechoslovakia, East/West Germany, North/South Yemen, Serbia–Montenegro, Netherlands Antilles; 149 events) and 2 territories (Azores, Canary Is.)	kept in aggregates; assigned to the region of the successor or administering state; excluded from country-level comparisons
No geography	no region or continent field	UN M49 sub-region (22) and continent added from ISO3 codes
Names	UN long forms; two "Korea" and two "Congo" after shortening	unique short display names per ISO3 code
Small countries	many entities with only a handful of events	country-level analyses use the 167 of 216 current entities with ≥ 10 events in 1970–2022

E. Derived network, multivariate, hierarchical and time-series data

The A1 table is not itself a network, a hierarchy or a multivariate table of comparable units, so each InfoVis task needed a remodelled dataset (Table IV). The network links countries whose mix of hazards is similar. The multivariate table gives one profile row per country, or per country and decade. The hierarchy rolls the table up into a decade × country × subtype cube from which several trees can be built. The time series are annual (or binned) totals and record-level distributions. Because the source has no month field, the required temporal variations use yearly vs. 5-year and decadal granularity, as well as different groups.

TABLE IV
DATASETS DERIVED FROM THE A1 TABLE FOR THE INFOVIS TASKS

Task	*Derived dataset*	*Size*
Node-link (Sec. VI)	country nodes with a 15-category hazard-share profile; edges between countries with similar profiles (Bhattacharyya coefficient, 4 nearest neighbours ≥ 0.70, plus maximum spanning tree)	167 nodes, 474 edges
Parallel coordinates (Sec. VII)	12 dimensions: events, deaths, affected, damage, deaths/event, 4 subgroup shares, number of hazard types, growth 1996–2022 vs. 1970–1995, share of records with damage reported; k-means clusters	167 countries; 622 country-decades
Treemap (Sec. VII)	rollup decade × continent × sub-region × country × subgroup × type × subtype with 4 measures, remodelled into 4 trees in the browser	4,937 leaf records
Horizon / ridgeline (Sec. VIII)	annual events by type (1950–2022), annual deaths by continent (1970–2022), 5-year rates, record-level distributions by type, decade and continent	10 + 5 + 10 series; ≈ 10 k records

III. SCIVIS TASK 1: COLOUR MAPPING *(LAKSHYA JAIN)*

A colour map assigns each scalar value a colour through a normalisation $n(x) = (x - a)/(b - a)$ followed by a palette lookup. Our experiments vary the palette, the transfer function (linear, logarithmic, discrete), the midpoint (sequential vs. diverging) and the parameters a, b (global vs. local minimum and maximum) [9]–[13].

A. Palettes

Fig. 2 maps the heat-wave day t3 (21 Jul) with four palettes on the same 9–49 °C range. Panel (e) shows each palette's CIE L* lightness profile. Viridis rises monotonically from L* ≈ 15 to 91 and YlOrRd falls monotonically from 99 to 26, so equal steps in data look like equal steps in colour. Both show the hottest areas (the Mojave and Sonoran deserts and the Central Valley above 45 °C), the cooler mountain ranges as thin lines (the lapse rate makes peaks 10–15 °C cooler) and the heat-wave footprint along the Mid-Atlantic and north-east (32–38 °C). YlOrRd has an intuitive "heat" connotation, but most of the country then falls into saturated oranges and reds, so the 30–38 °C range that carries the heat-wave signal has little contrast. Jet reaches L* ≈ 90 at 0.38 of its range, stays nearly flat until 0.63 and then falls. Its bright cyan and yellow bands create false boundaries near 22 °C and 35 °C that look like fronts but are artefacts of the palette [9]; it is also not safe for colour-blind readers or greyscale printing. The diverging RdYlBu_r palette centred at 35 °C (d) is the only one here that answers a threshold question, which areas exceed a heat-stress level. That is legitimate only because 35 °C is a meaningful reference [10]. A diverging map around an arbitrary centre would split the data into two halves that do not mean anything.

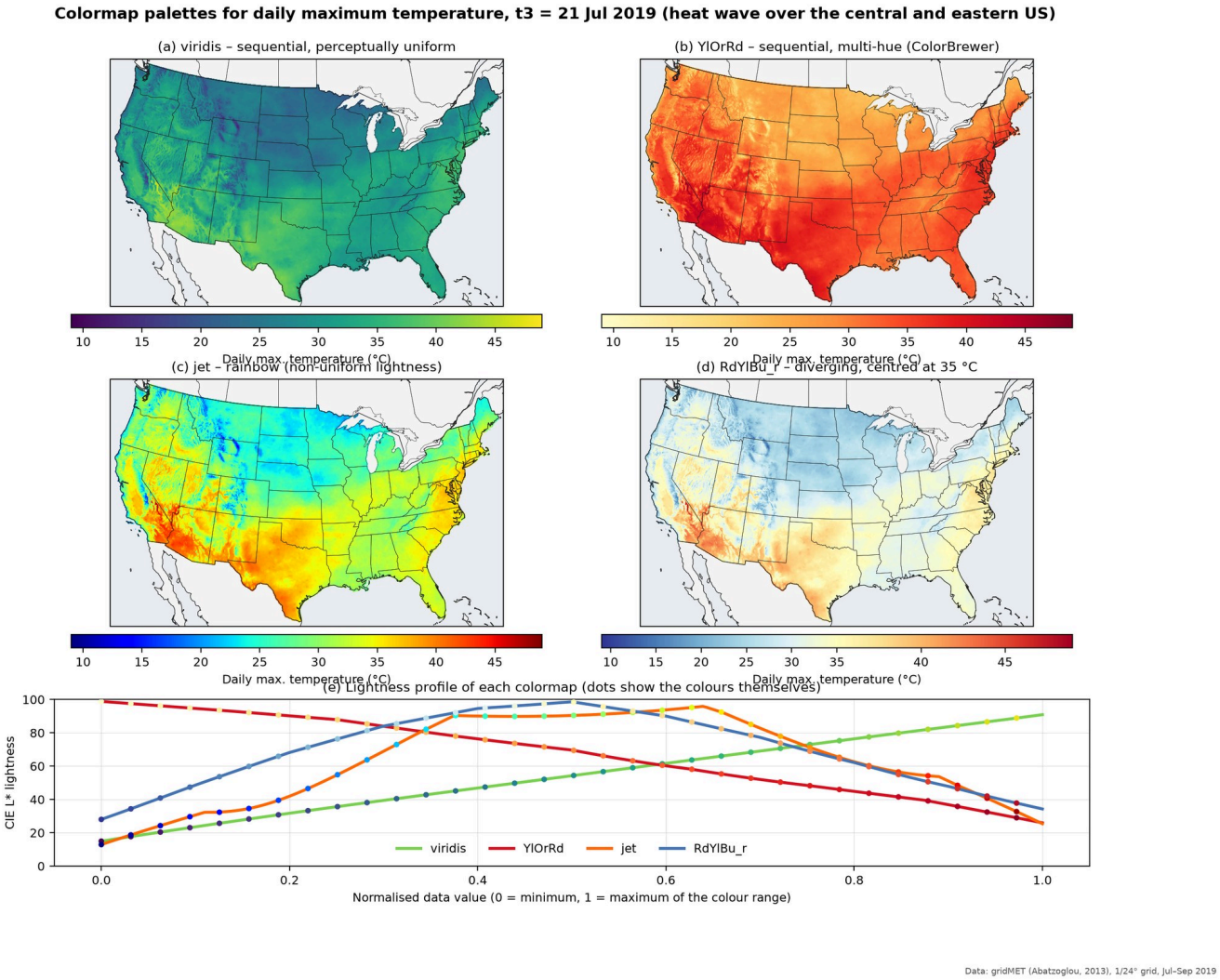


Fig. 2. Daily maximum temperature on t3 = 21 Jul 2019 with (a) viridis, (b) YlOrRd, (c) jet and (d) RdYlBu_r centred at 35 °C; (e) CIE L* lightness of each colormap along its range (dots are the colours themselves).

B. Continuous vs. discrete mapping

Fig. 3 compares a continuous palette with two discrete classifications on t10 (29 Sep), the day with the strongest north–south contrast. With equal-interval classes (11 classes of 5 °C), the cold outbreak is a compact, sharply bounded region: cells below 10 °C cover 19.9 % of the land, in Montana, Wyoming and Idaho. The class above 35 °C covers only 4.1 %, in Texas and the Southeast. Quantile classes (8 classes of equal area) have bounds at −11.5, 6.6, 12.9, 18.5, 23.2, 29.2, 31.9, 33.8 and 38.6 °C. The lowest class alone spans 18 °C, while the three warmest classes together span 9.4 °C. Quantiles therefore exaggerate small differences across the warm South and Plains and hide the magnitude of the cold anomaly. Equal-interval classes are preferable when absolute thresholds such as freezing or heat stress matter, and a quantile map is really a rank map. Discrete classes make values easy to read off the legend; the continuous map keeps all the detail but makes exact reading harder.

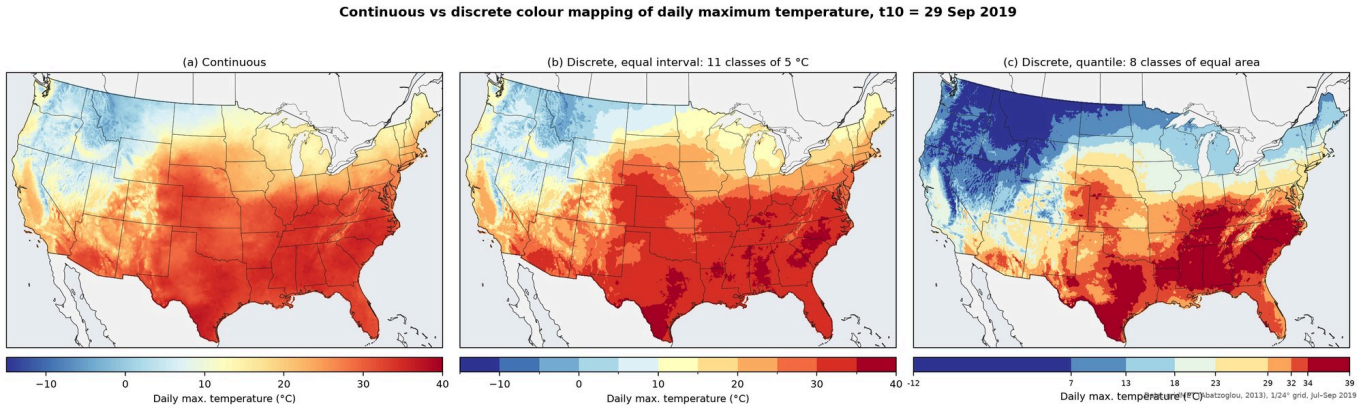


Fig. 3. Continuous mapping (a) vs. discrete equal-interval (b) and quantile (c) classes of daily maximum temperature on t10 = 29 Sep 2019.

C. Global vs. local minima and maxima across time steps

With **global** normalisation (Fig. 4; GIF `Fig04_…_animation.gif`), one scale from −11.5 °C to 49.5 °C (the extremes over all ten samples) is used for every frame. Colours then mean the same temperature in every panel, so the seasonal cooling is visible as a progressive darkening. The northern tier darkens first (t8, 9 Sep), the Great Basin follows on t9, and the Montana–Wyoming cold outbreak on t10 is the darkest area of the whole sequence. With **local** normalisation (Fig. 5; GIF `Fig05_…`), each frame is stretched to its own range, from 9.4–48.1 °C on t3 to −11.5–38.6 °C on t10. Within-day contrast is maximised, which helps the reader see each day's spatial pattern. Across days, however, colours lose their meaning: the Southeast on t10 (34 °C) looks as "hot" as the 46 °C deserts on t1, and the animation flickers because the scale changes rather than the temperature. Fig. 6 quantifies the trade-off: each day uses only 61–82 % of the global range, and the global extremes are set by a handful of cells (desert floors near 49 °C, high peaks on t10 at −11.5 °C). A robust global range from the 1st to the 99th percentile (5.2–40.2 °C) would add contrast and keep values comparable over time, at the cost of saturating 2 % of the cells. We therefore recommend global or robust-global normalisation for any temporal comparison or animation, and local normalisation only for single-day exploration, always with a separate colour bar per frame.

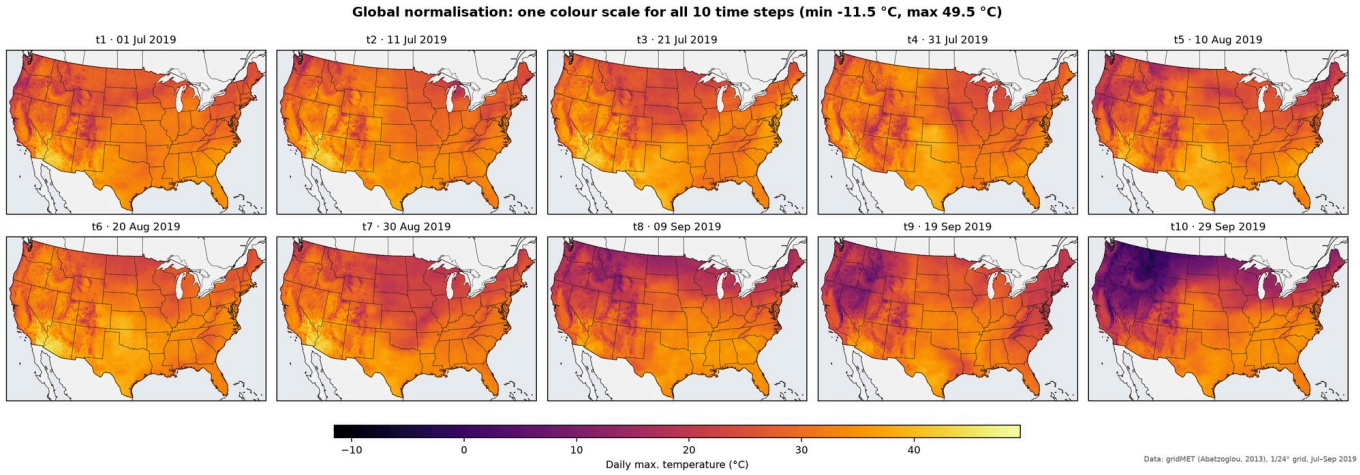


Fig. 4. Global normalisation: daily maximum temperature on all ten time steps with one colour scale (inferno, −11.5 to 49.5 °C). Animation: `images/Fig04_tmmx_global_minmax_animation.gif`.

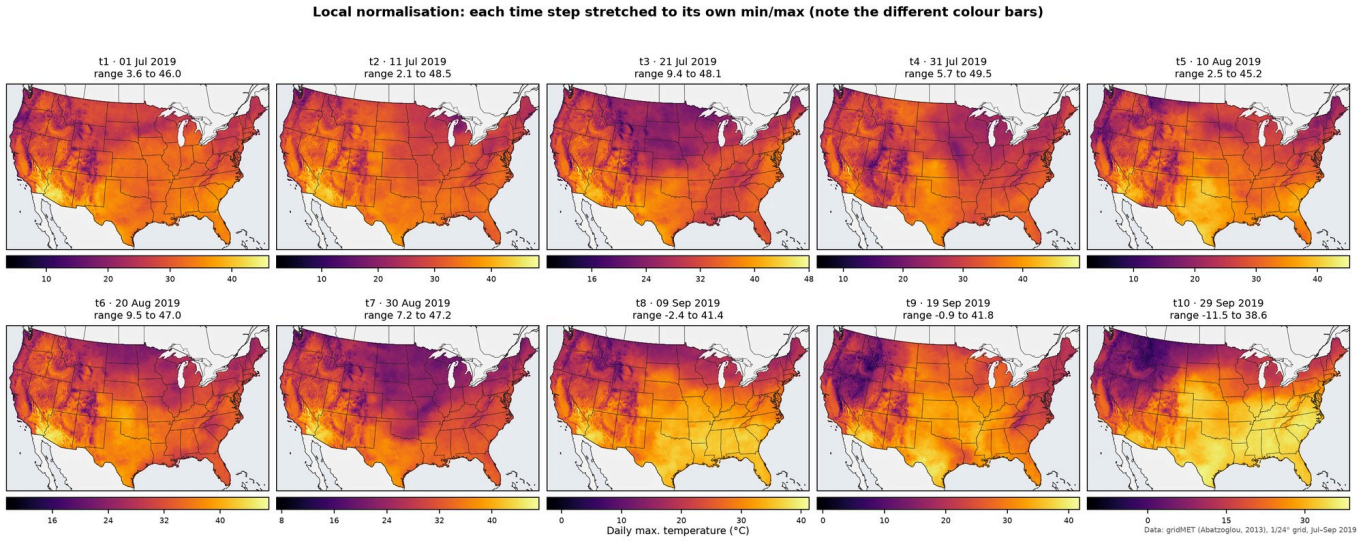


Fig. 5. Local normalisation: each time step stretched to its own minimum and maximum (range printed in each title; note the different colour bars). Animation: `images/Fig05_tmmx_local_minmax_animation.gif`.

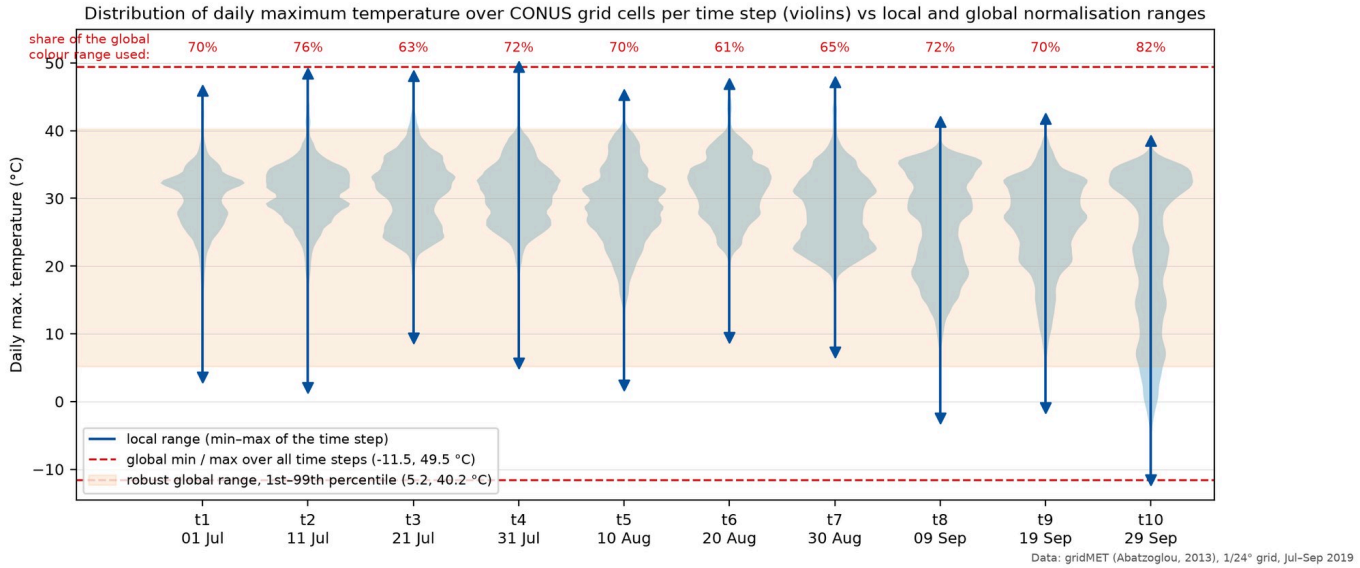


Fig. 6. Distribution of grid-cell values per time step (violins), local ranges (arrows), global range (dashed) and robust global range (1st–99th percentile, shaded); the percentages give the share of the global colour range that each day uses.

D. Diverging colour map for anomalies

Absolute temperature has no natural midpoint, but an anomaly does: zero. Fig. 7 (GIF `Fig07_…`) maps each sample's departure from the ten-sample mean with RdBu_r centred at 0 and symmetric limits of ±21 °C (the 99.5th percentile of |anomaly|). The diverging map reveals what neither sequential map can show. 31 Jul (t4) was up to +13 °C warmer than normal over Montana and the Dakotas. 20 Aug (t6) was warm over the North-west. The first half of September cooled the whole northern half of the country, and on 29 Sep (t10) Montana and Wyoming were up to 25 °C below their quarter mean while the Southeast remained 2–5 °C above it. The same 25 °C means “warm” in Montana and “cool” in Texas, so only the anomaly map with a neutral white for “normal” lets one colour carry the same meaning everywhere.

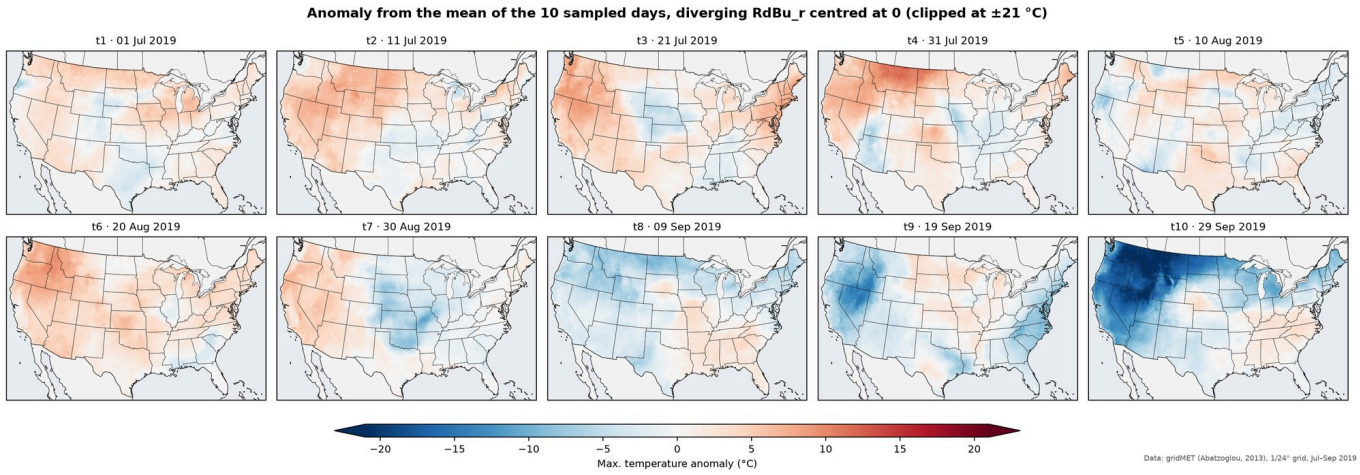


Fig. 7. Daily maximum-temperature anomaly from the mean of the ten sampled days, diverging RdBu_r centred at 0, clipped at ±21 °C. Animation: `images/Fig07_tmmx_anomaly_diverging_animation.gif`.

E. Logarithmic and discrete mapping of precipitation

Precipitation is zero-inflated and heavy-tailed. On t9 (19 Sep), 66.9 % of land cells were dry (< 0.1 mm), and the wet cells had a median of 2.2 mm, a 90th percentile of 15.7 mm, a 99th percentile of 61.2 mm and a maximum of 295 mm (Fig. 8d). A linear map (Fig. 8a) squeezes 99 % of the wet cells into the lowest fifth of the palette, so the map shows only Imelda's core between Houston and Lake Charles. A logarithmic map (b), with dry cells left neutral grey because log 0 is undefined, reveals the whole precipitation field. It shows a frontal rain band from the Upper Midwest to Texas, widespread rain over Oklahoma, Arkansas and Louisiana around Imelda's remnant circulation, showers in the northern Rockies and Great Basin, and convection over Florida. A multiplicative, roughly log-normal quantity is perceived correctly only on a log scale. Discrete rainfall-intensity classes (c) make values readable, e.g. “more than 200 mm” around Beaumont–Port Arthur, at the cost of within-class detail. Fig. 9 (GIF `Fig09_…`) applies one global log scale to all ten days. The wettest days by area were t10 (38 % of land ≥ 1 mm, a rain shield across the northern tier ahead of the cold outbreak) and t5 (37 %); the West stays mostly dry apart from monsoon showers over Arizona, New Mexico and Colorado on t4–t5.

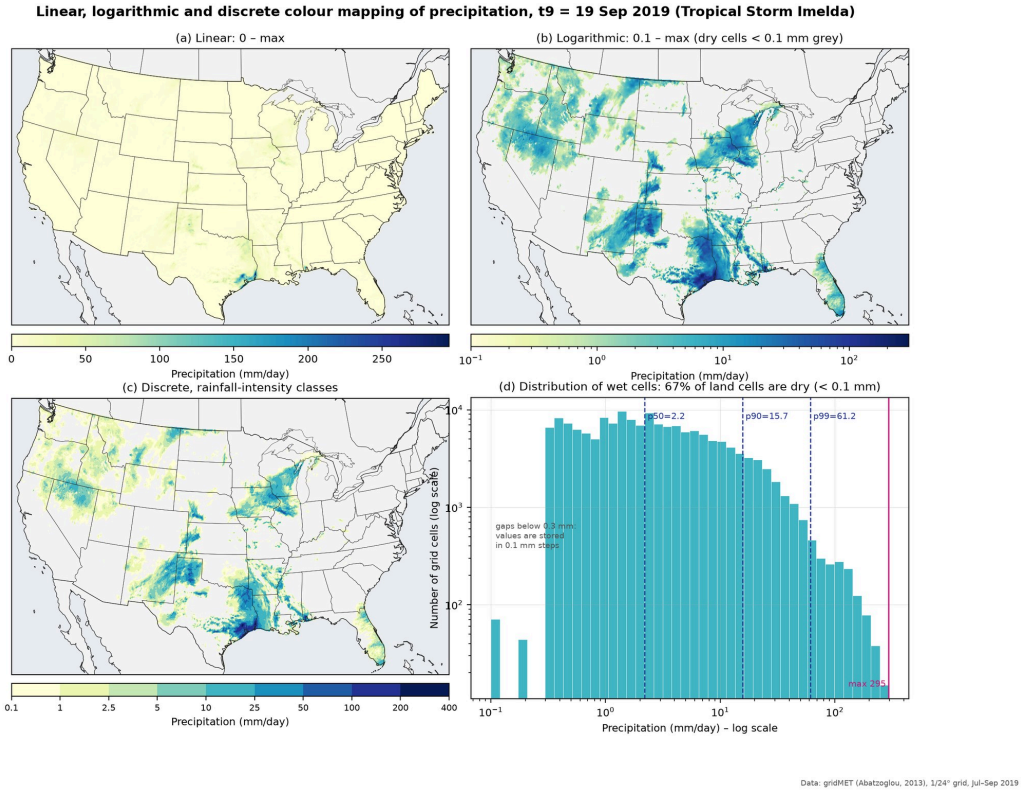


Fig. 8. Precipitation on t9 = 19 Sep 2019 (Tropical Storm Imelda): (a) linear, (b) logarithmic and (c) discrete colour mapping (YlGnBu); (d) log–log histogram of wet cells with percentiles (gaps below 0.3 mm reflect the 0.1 mm storage step).

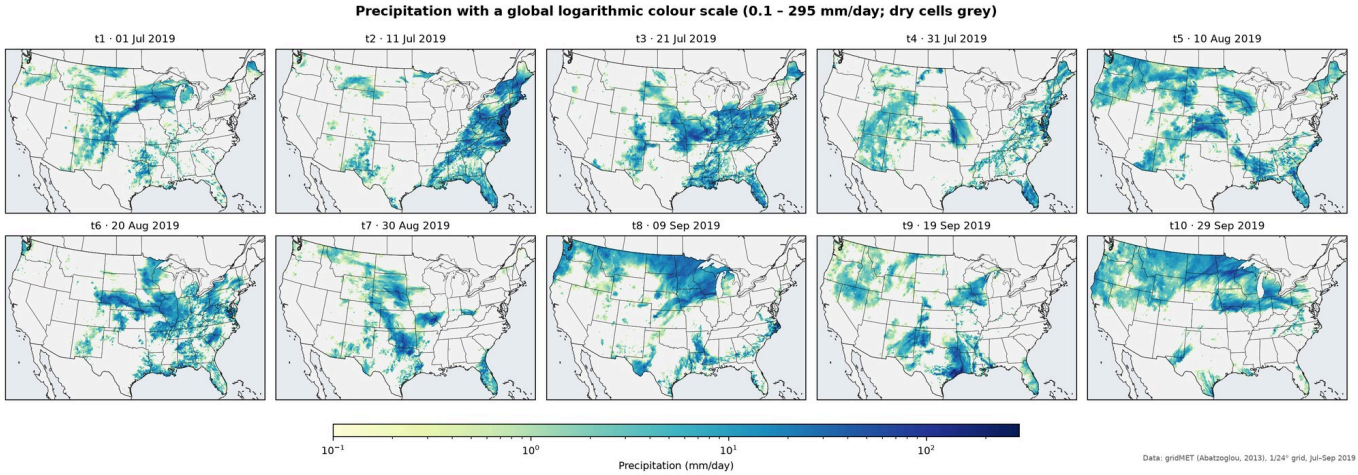


Fig. 9. Precipitation on all ten time steps with a global logarithmic colour scale (0.1–295 mm/day). Animation: `images/Fig09_precip_global_log_animation.gif`.

IV. SciVis Task 2: Contour Mapping *(Kalpit)*

A. Choice of field and algorithm

Contours were computed for vapour pressure deficit (VPD), the difference between saturation and actual vapour pressure. VPD is smooth at synoptic scales and has physically meaningful thresholds: below about 1 kPa the air is humid, and above about 3 kPa it is very dry, with high evaporative demand, plant water stress and fire weather. It is also independent of the colour-mapping variables. We use **both** permitted algorithms, each for what it does best. **Marching Squares** extracts isolines: each grid cell is classified independently by which of its four corners lie above the isovalue, and the crossing points are placed on the cell edges by linear interpolation [14]. The method is local, so cells with a missing (ocean) corner are simply skipped. It achieves sub-cell accuracy, has a deterministic rule for ambiguous saddle cells, and runs in O(N) time, which makes it ideal for precise threshold boundaries such as the 1 kPa and 3 kPa lines. **Contour fill** (matplotlib's contourf [6], [15]) builds filled polygons between consecutive levels. It answers areal questions ("where is VPD between 2 and 3 kPa, and how large is that region?") and acts as a discretised colour map whose class boundaries are the isolines.

B. Implementation and verification

Our implementation (`scivis/marching_squares.py`) is fully vectorised in NumPy. For a cell with corners a (top-left), b (top-right), c (bottom-right) and d (bottom-left), the case index is $8[a \geq c_0] + 4[b \geq c_0] + 2[c \geq c_0] + [d \geq c_0]$, which selects one of 16 cases (Fig. 10a). Each crossing lies at t =

$(c_0 - p)/(q - p)$ along an edge pq. The saddle cases 5 and 10 are resolved with the cell-centre value $(a + b + c + d)/4$. The data revealed one subtle issue. VPD is stored in steps of 0.01 kPa, so 3,303 cells on 21 Jul equal exactly 1.00 kPa. An isoline drawn exactly at a stored value depends on whether "equal" counts as above or below: our ≥ rule and contourpy's > rule then differ by up to 1.34°, because flat plateaus at exactly the level are bounded on different sides. We therefore evaluate each nominal level L at $L - 0.005$ kPa, which is unambiguous because it lies between two storable values. With this choice our isolines match contourpy's to within 2.9×10^{-14}° for all levels, i.e. they are identical up to floating-point rounding (Table V). Linear interpolation matters as well. Placing crossings at edge midpoints (Fig. 10b, orange) produces a staircase offset by up to half a cell (2 km), whereas the interpolated line (red) follows the reference exactly. In the worked example the 3 kPa isoline separates the dry Salt Lake valley from the moister Wasatch Range.

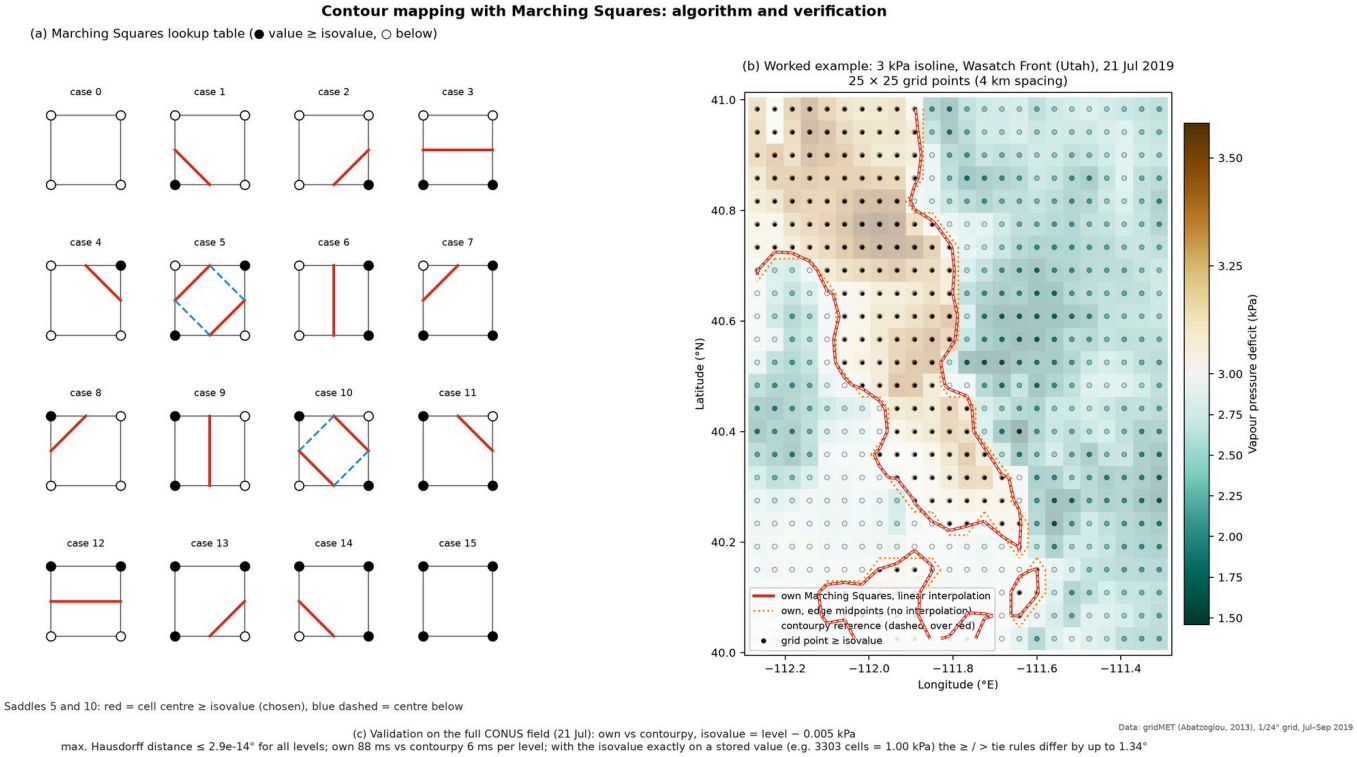


Fig. 10. (a) The 16-case Marching Squares lookup table, with both resolutions of the saddle cases. (b) Worked example: the 3 kPa isoline on a 25 × 25 sub-grid over the Wasatch Front (21 Jul) from our implementation with linear interpolation (red), with edge midpoints (orange) and from contourpy (white dashed). (c) Validation summary over the full CONUS field.

TABLE V
OWN MARCHING SQUARES VS. CONTOURPY, VPD ON 21 JUL 2019

Level (kPa)	*Segments*	*Length (km)*	*Saddles*	*Own / lib (ms)*	*Hausdorff (°)*
0.5	7,800	22,760	144	79 / 6	1.6e−14
1.0	20,873	60,749	539	96 / 9	2.9e−14
2.0	14,175	42,459	237	96 / 7	2.9e−14
3.0	9,376	29,634	91	82 / 5	2.8e−14
4.0	3,720	11,629	72	87 / 5	2.8e−14

C. Isolines, line contours and filled contours

Fig. 11 shows the five isolines on 21 Jul. The 4 kPa isoline encloses the Mojave and Sonoran deserts and the southern Central Valley. The 3 kPa isoline extends over the Great Basin and into west Texas. The 1 kPa isoline traces the edge of the humid East and wraps around nearly every mountain range in the West as thousands of small closed loops. That makes it the longest isoline (60,749 km), even though about 38 % of the land lies below it. The length of an isoline therefore measures how fragmented a threshold boundary is, not how large the enclosed area is. Fig. 12 compares the two contour techniques. Isolines alone (a) show exactly where the thresholds lie, but the reader must infer which side is higher, and the mountain loops add clutter. Five filled bands (b) make regions and their areas readable at a glance, but they hide all variation inside the wide 1–2 kPa band that covers much of the Plains. Ten filled levels with the isolines overlaid (c) combine both strengths and show the stepwise gradient from the humid Gulf coast to the dry Southwest. The continuous map (d) keeps every detail, but no threshold can be read from it.

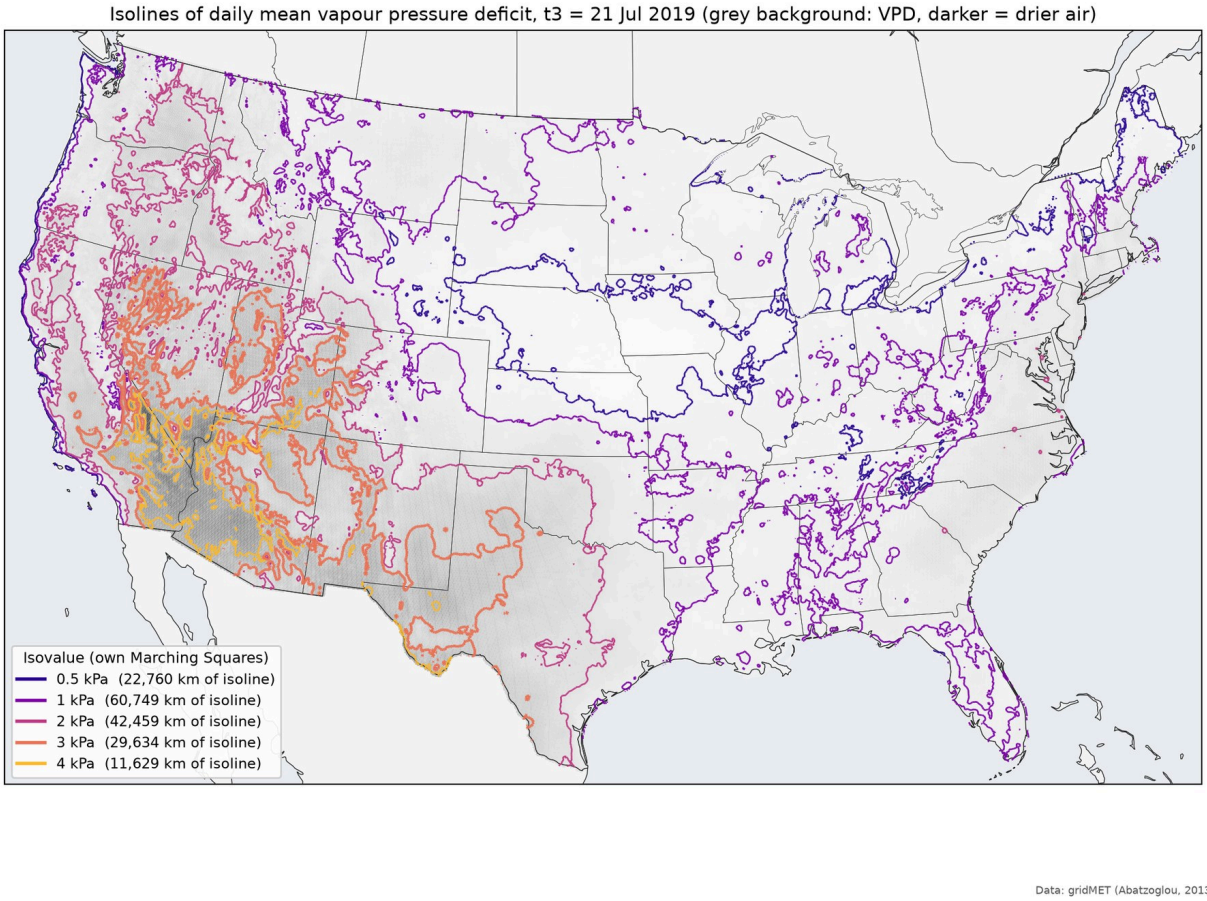


Fig. 11. Isolines of daily mean vapour pressure deficit on t3 = 21 Jul 2019 from our Marching Squares implementation (0.5–4 kPa) over a grey VPD background; the legend gives the total length of each isoline.

D. Effect of pre-smoothing

At 4 km, VPD follows the terrain, so raw isolines in the West are dominated by small, closed loops. Fig. 13 applies a mask-aware Gaussian filter before contouring. It uses normalised convolution, so values near the coast are not pulled towards zero by the missing ocean cells. Smoothing with $\sigma = 2$ cells (8 km) shortens the total length of the 1–4 kPa isolines over CONUS from 144,470 km to 65,379 km (−55 %) and cuts the number of saddle cells at 1 kPa from 539 to 3, while the large-scale boundaries barely move. With $\sigma = 5$ cells (20 km) the length falls to 39,999 km (−72 %), isolated basins disappear and neighbouring dry regions merge. For synoptic interpretation $\sigma \approx 2$ cells is a good compromise. Either way, smoothing alters the data and must be disclosed alongside the result.

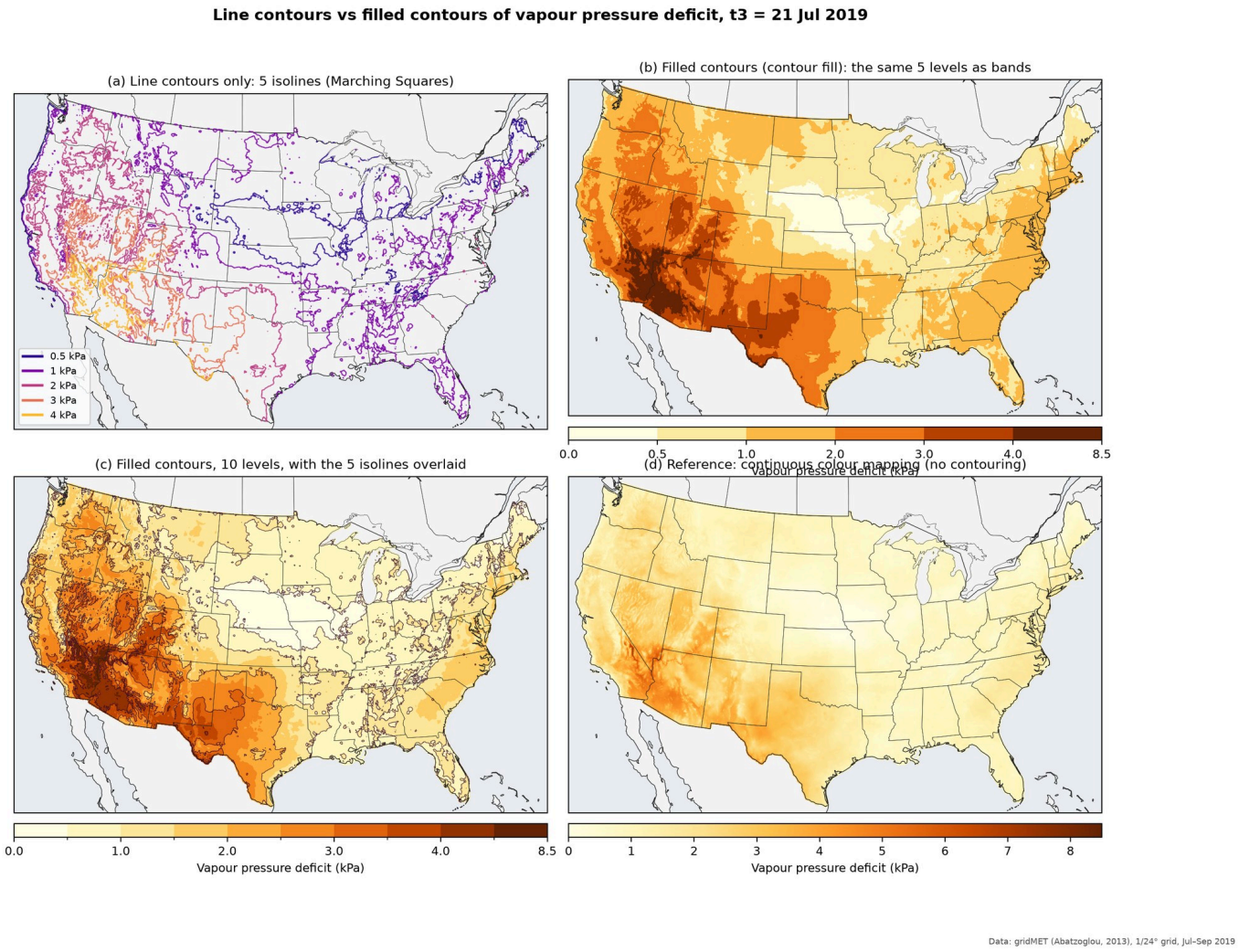


Fig. 12. Line contours vs. filled contours of VPD on 21 Jul 2019: (a) five isolines, (b) filled contours with the same five levels, (c) ten filled levels with the isolines overlaid, (d) continuous colour mapping for reference.

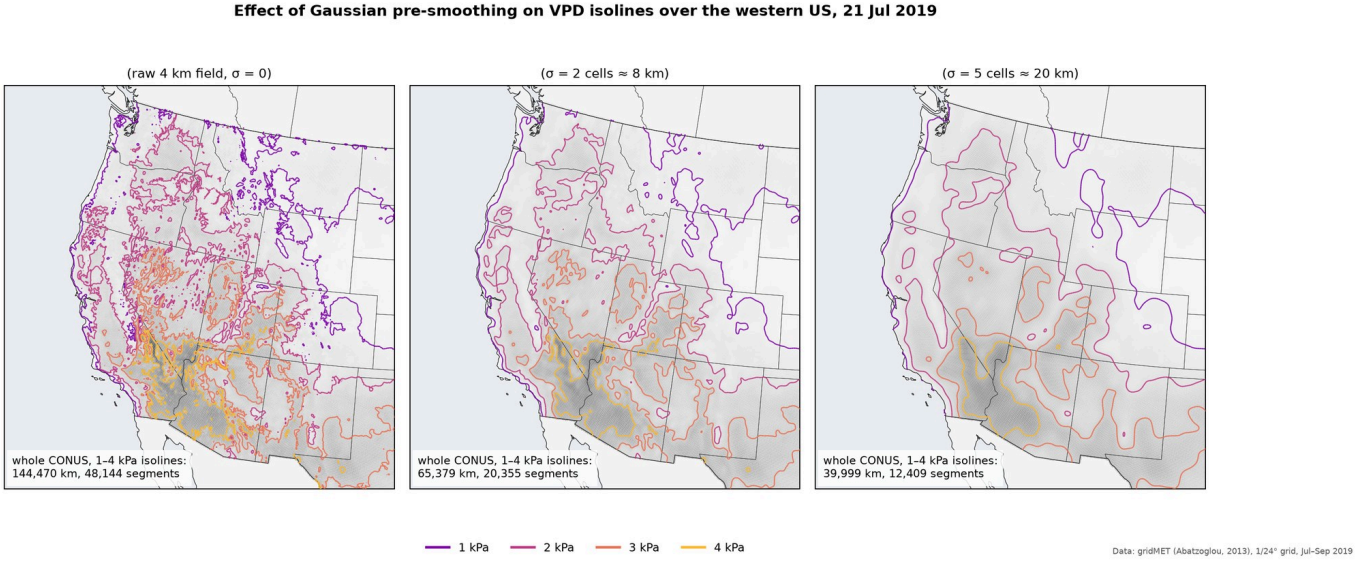


Fig. 13. VPD isolines over the western US on 21 Jul 2019 for the raw field and after Gaussian smoothing with σ = 2 and 5 cells; the boxes give the CONUS-wide total isoline length and segment count.

E. Evolution over the ten time steps

Fig. 14 (GIF `Fig14_…`) uses fixed levels on all ten days, and Fig. 15 summarises the contours quantitatively. The very dry region (VPD ≥ 3 kPa) is largest on t3 (0.76 M km², 9.6 % of the land) and t6 (0.68 M km²), when it covers the desert Southwest and Great Basin and, on t4–t5, extends over the southern Plains. In September it collapses: to 0.07 M km² on t8 and 0.12 M km² on t9, and on t10 no grid cell reaches 3 kPa at all. The area with VPD ≥ 2 kPa falls from 33 % of the land (t6) to 3 % (t10), and the area with VPD ≥ 1 kPa from 73 % (t1) to 44 % (t10). The ten sampled points (Fig. 15a)

follow the daily 16 km series closely, which confirms that the uniform samples represent the quarter. The daily series also shows day-to-day swings of several percentage points that the samples cannot resolve.

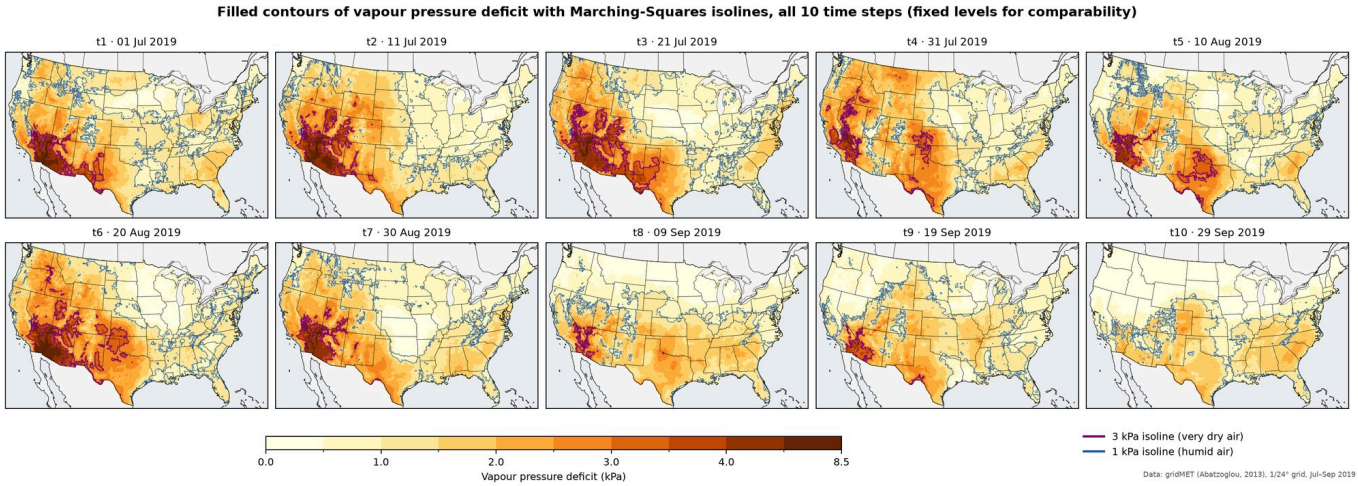


Fig. 14. Filled VPD contours (fixed levels) with the 1 kPa (blue) and 3 kPa (purple) Marching-Squares isolines for all ten time steps. Animation: `images/Fig14_vpd_contours_animation.gif`.

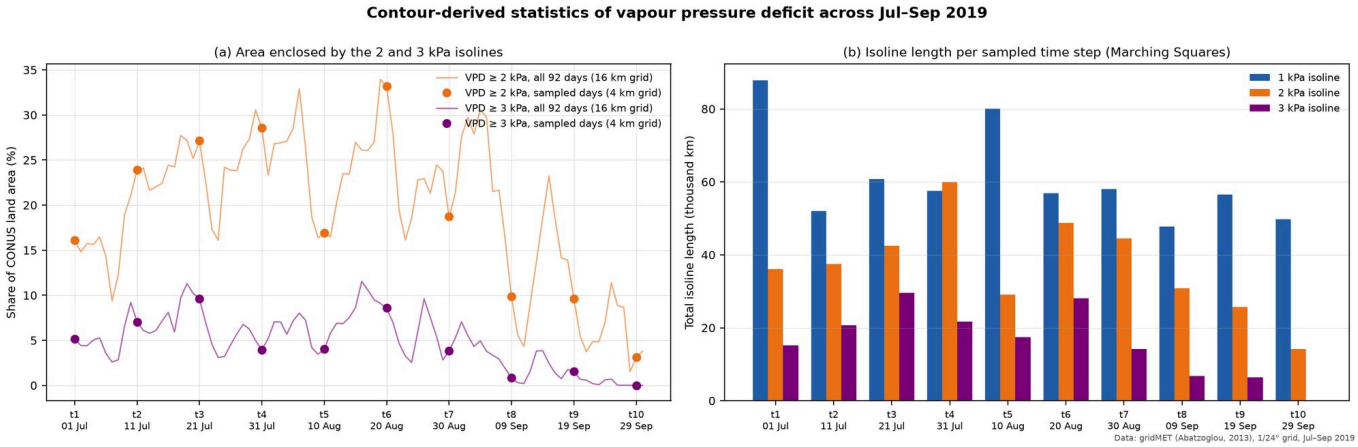


Fig. 15. Contour-derived statistics: (a) share of land enclosed by the 2 and 3 kPa isolines on all 92 days (lines, 16 km grid) and on the sampled days (dots, 4 km grid); (b) total isoline length per sampled day.

V. SciVis Task 3: Quiver Plots and Streamlines *(Arnav)*

A. The wind field

Vectors are reconstructed from *vs* and *th* with (1). Two properties of the data shape every experiment. First, *vs* and *th* are distributed as separate daily aggregates, so the reconstructed vector only approximates the daily mean wind; on days with turning winds the true vector mean is shorter than *vs*. Second, gridMET's wind inherits the ≈12 km (1/8°) structure of the NLDAS-2 forcing it is derived from [1], [19], and wind is available over land only, so offshore circulations such as hurricanes are not visible. Over the quarter the mean speed was 3.0–3.5 m/s in July and August and rose to 4.0 m/s on t8–t9 and 5.2 m/s on t10 (95th percentile 8.9 m/s; maximum 16.0 m/s over north-west Colorado). The ratio of the length of the CONUS-mean vector to the mean speed is only 0.13–0.35 on every day, so there is never a single prevailing flow direction. This is why maps, rather than averages, are needed.

B. Grid-sampling experiments

A 4 km grid has 0.8 M vectors, far too many to draw, so the field must be sampled. In Fig. 16 the arrow length is tied to the sampling interval (a 10 m/s arrow spans one interval), which keeps the density of ink comparable across panels. On t10 every 10th cell (a, 4,816 arrows, 40 km) turns the field into texture whose arrows are too short to read. Every 20th cell (b, 1,230 arrows, 80 km) gives the best balance and shows the configuration of the early-season storm. Strong south-westerlies cross the Great Basin and the Four Corners ahead of a deep western trough; a southerly flow from the Gulf covers Texas, Oklahoma and Kansas; and easterly upslope winds blow behind the cold front over Montana and the Dakotas. Every 40th cell (c, 305 arrows) represents the wind shift across the Dakotas with only two or three arrows. Block-averaged vectors (d), the mean u and v over each 20 × 20 block, are nearly identical to point samples (b). The median direction difference is only 3°, 0.7 % of blocks differ by more than 45°, and within a block the vector-mean speed is 0.99 of the scalar-mean speed (f). Aliasing is negligible here because the field has almost no structure below 80 km. Block averaging remains the safer default for noisier, finer data. Jittered sampling (e, one random cell per block)

removes the regular grid pattern that the eye can mistake for structure, at the cost of small local gaps and clumps.

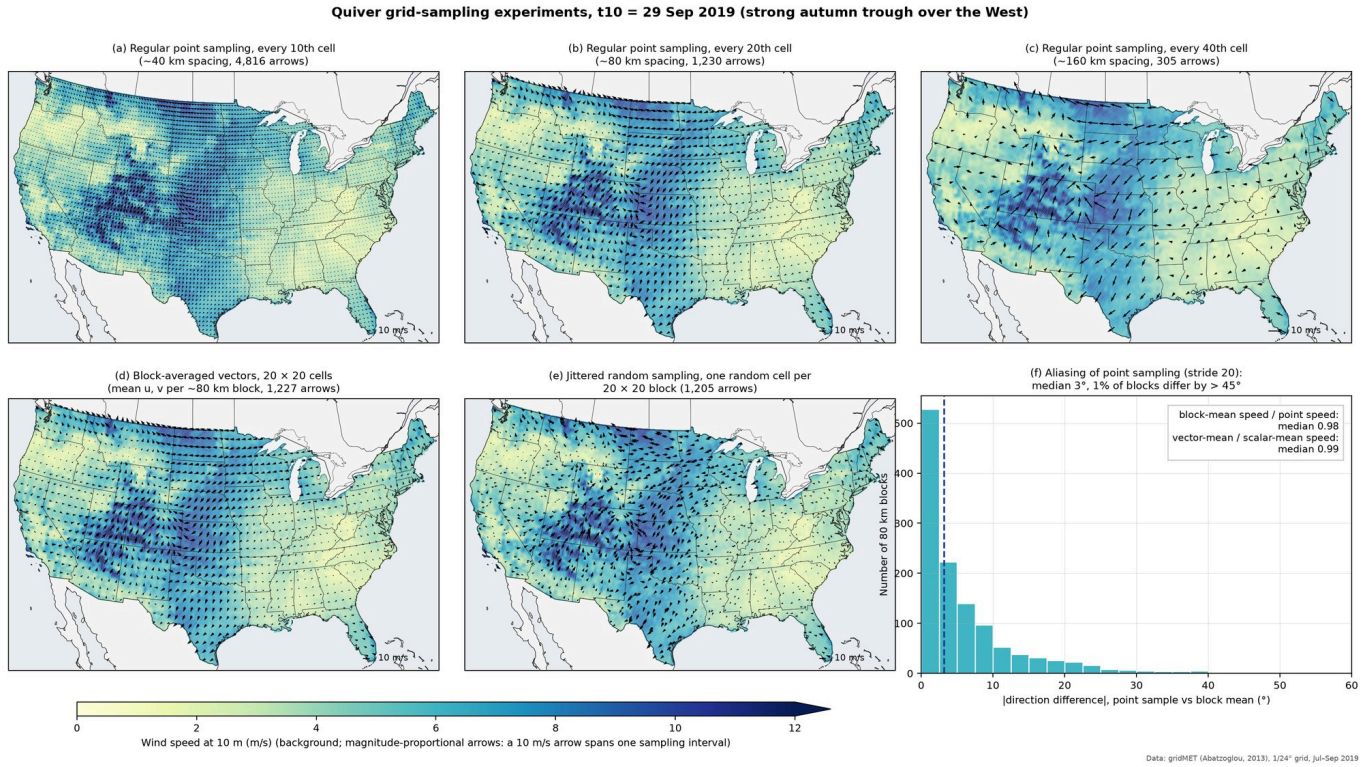


Fig. 16. Quiver grid-sampling experiments on t10 = 29 Sep 2019: regular point sampling every (a) 10th, (b) 20th and (c) 40th cell, (d) block-averaged vectors, (e) jittered random sampling, (f) direction difference between point and block-mean vectors.

C. Magnitude-proportional vs. same-sized vectors

Magnitude-proportional arrows (Fig. 17a) emphasise the strong-wind corridors over the Great Basin and High Plains. In light-wind regions such as the Southeast and the Pacific coast they shrink to dots whose direction cannot be read. Same-size arrows (b) show direction everywhere and encode speed by colour. They are better for reading circulation patterns, such as the contrast between south-westerly and easterly flow on either side of the front. Comparing magnitudes is harder, though, because colour is judged less accurately than length. Square-root scaling (c) compresses the 1–16 m/s range into a factor of four, so light winds remain visible and strong winds are still longer. It is a good compromise when both direction and magnitude matter.

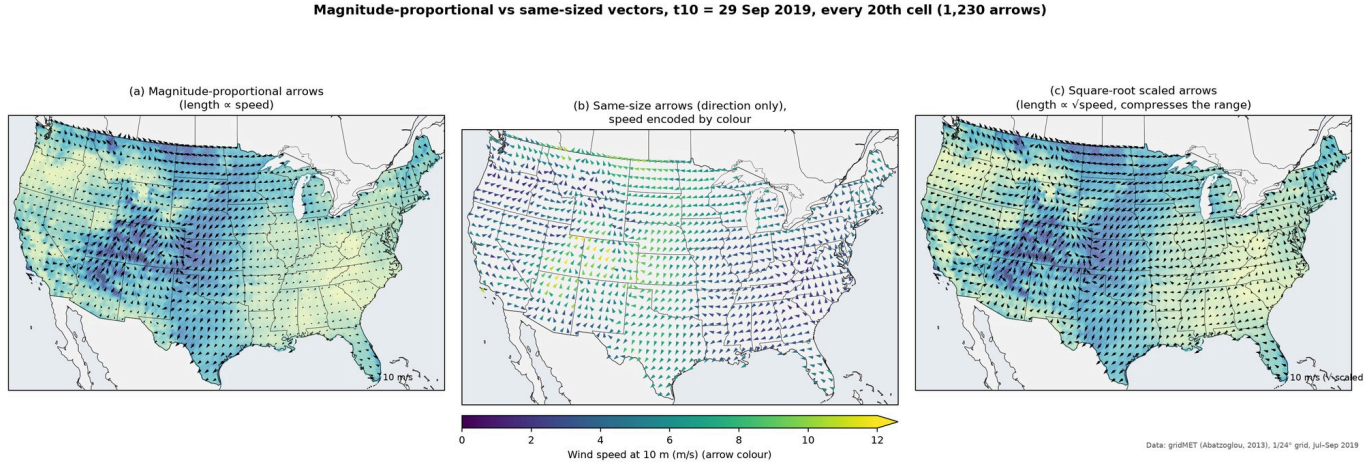


Fig. 17. (a) Magnitude-proportional, (b) same-size (direction-only, speed by colour) and (c) square-root-scaled arrows on t10, every 20th cell.

D. Streamline seeding strategies

Streamlines are integral curves of the field. Their appearance depends on where integration starts (the seeds) and on when it stops, at the collision density that keeps lines apart [16], [17]. Fig. 18 and Table VI compare four strategies on t10 with the same collision density, so only the seeds differ. Density-based seeding, matplotlib's evenly-spaced placement, tries seeds in every empty cell. It produces 106 streamlines that cover 56 % of the 60 km land cells, the most uniform picture. Uniform-grid and random seeding with 126 seeds each produce only 52 and 56 streamlines (39 % coverage each). Many seeds fall into cells already crossed by a streamline and are rejected, and random seeds leave visible

gaps and clusters. Feature-based seeding, with seeds drawn from the fastest 5 % of the field (≥ 8.9 m/s), deliberately ignores calm regions. Its 21 streamlines trace exactly where the strong flow goes: from the Great Basin into Wyoming, and up the Plains towards the Great Lakes.

TABLE VI
STREAMLINE SEEDING STRATEGIES, 29 SEP 2019

Strategy	*Seeds*	*Streamlines*	*Coverage*
Density-based (evenly spaced)	auto	106	55.7 %
Uniform grid	126	52	38.9 %
Random	126	56	38.8 %
Feature-based (top 5 % speed)	126	21	15.9 %

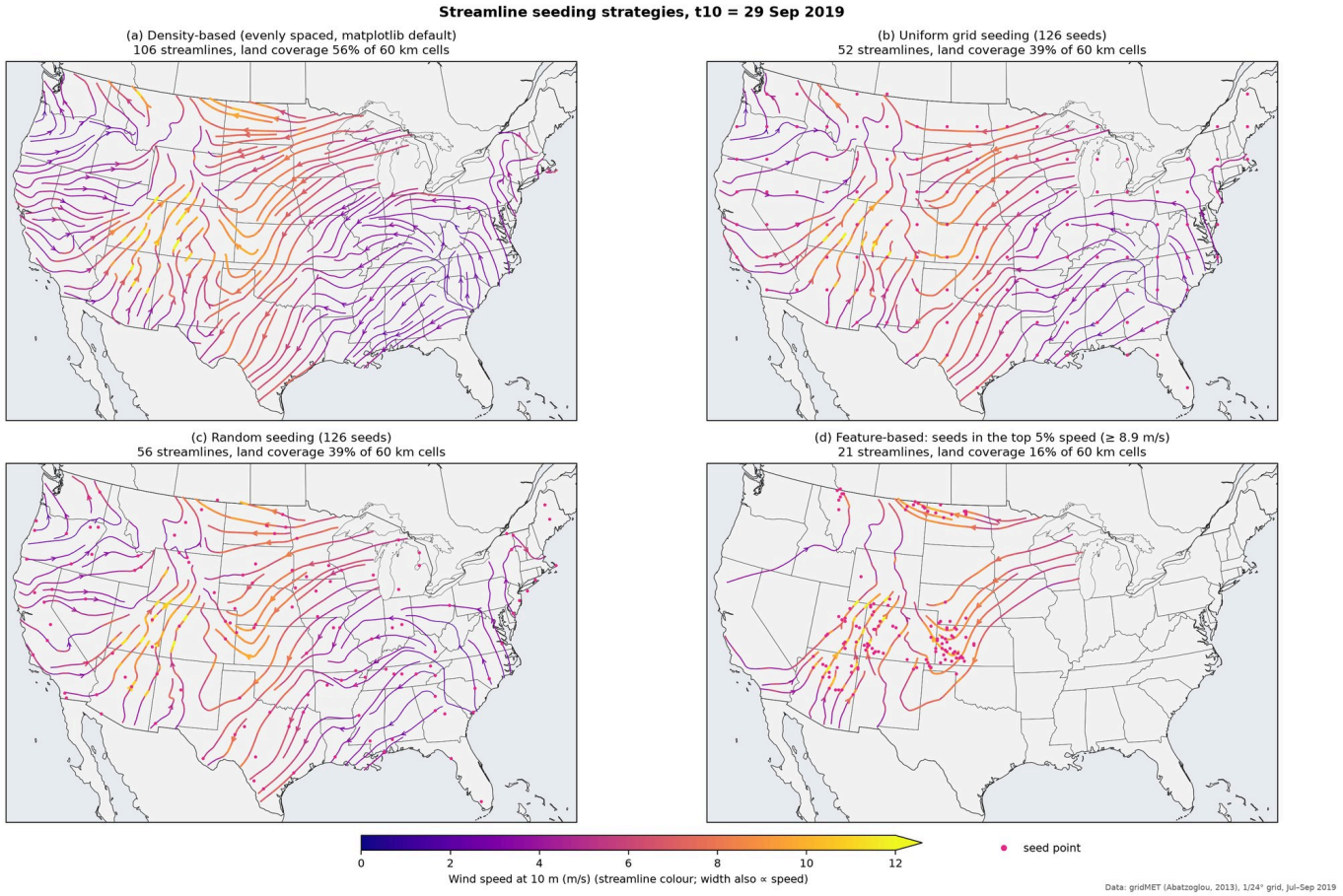


Fig. 18. Streamline seeding strategies on t10: (a) density-based, (b) uniform grid, (c) random and (d) feature-based seeds (pink dots); colour and width encode speed.

E. Quiver plots vs. streamlines

Fig. 19 contrasts the two methods in strong flow (t10, a zoom on the Central Rockies and High Plains) and in weak flow (t3, all of CONUS). Streamlines expose the topology of the flow that arrows only imply. On 29 Sep they show the confluence line where the south-westerlies over Utah meet the easterly upslope flow over Wyoming and Montana, roughly along the Continental Divide, together with the turning of the southerly Plains flow towards the north-east. On 21 Jul they show the large-scale circulation behind the weak arrows: south-westerly flow around the Atlantic subtropical high over the East, which fed warm, humid air into the heat wave, and northerly flow behind the cold front over the central Plains. Quiver plots, in turn, show the local magnitude at every sampled point without bias and need no integration. Streamlines lose magnitude unless it is encoded by width or colour, as in Fig. 18. They also suggest a continuous, physically consistent flow that a daily mean field does not strictly have. We therefore use streamlines for structure and quiver plots for local measurement, a division of labour that is consistent with user studies of 2D vector-field methods [18].

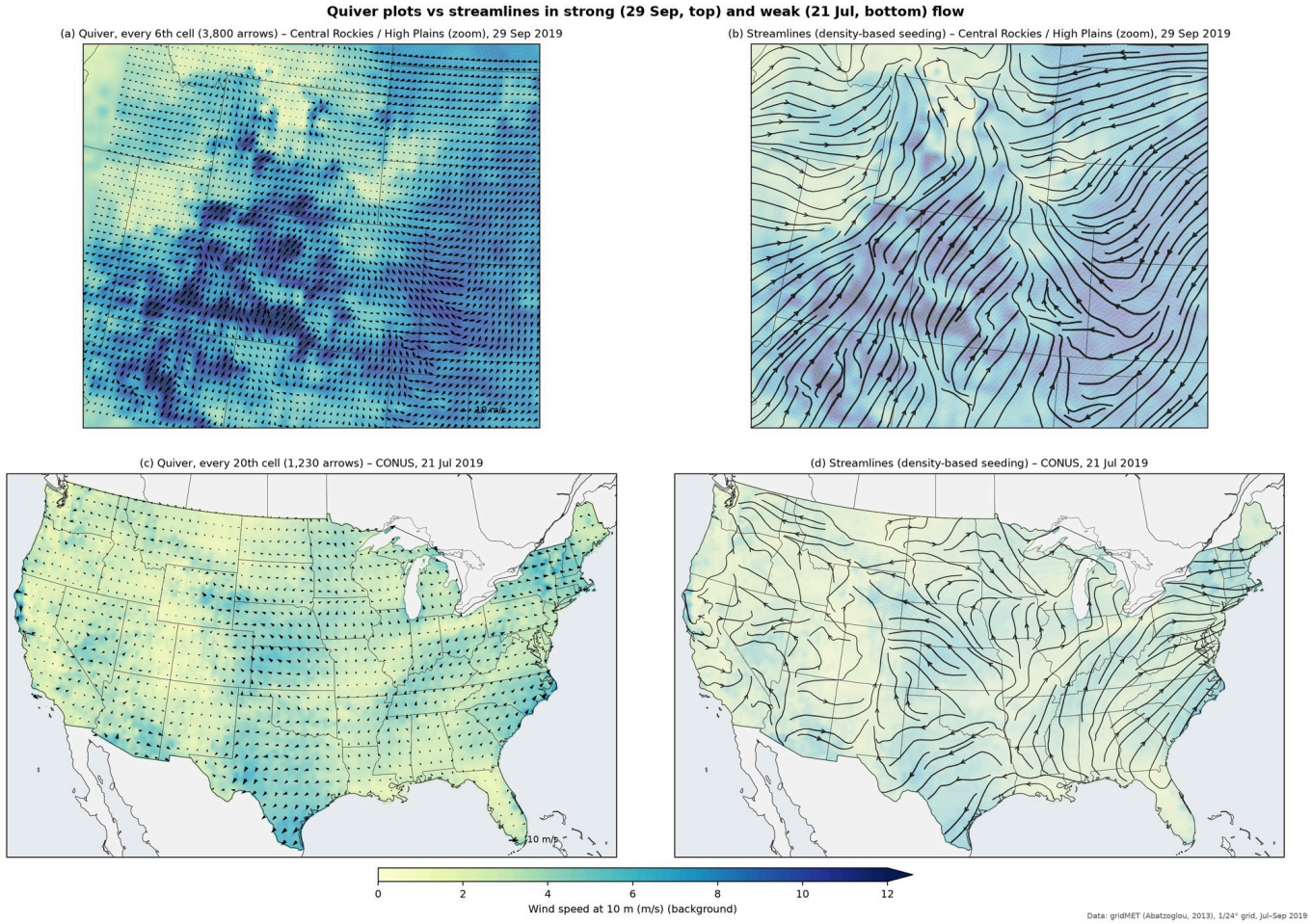


Fig. 19. Quiver plots (left) vs. streamlines (right) in strong flow (29 Sep, Central Rockies and High Plains, top) and weak flow (21 Jul, CONUS, bottom).

F. Evolution over the ten time steps

Figs. 20 and 21 (GIFs `Fig20_…`, `Fig21_…`) show the wind on all ten days. July and August are dominated by weak, disorganised flow (CONUS mean 3.0–3.5 m/s), with the persistent strong winds limited to the Pacific coast gaps and the southern Plains. From t8 the flow strengthens and organises around passing troughs, ending in the strong, spatially coherent pattern of t10, which is also the day with the largest anomalies in temperature (Fig. 7) and precipitation area (Fig. 9).

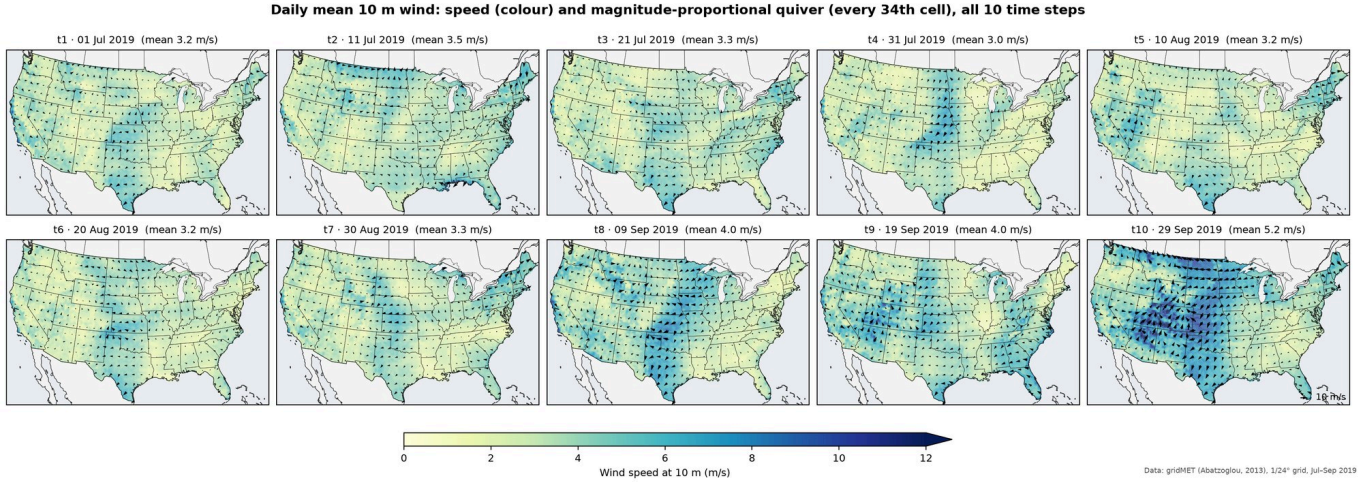


Fig. 20. Wind speed (colour) and magnitude-proportional quiver (every 34th cell) for all ten time steps. Animation: `images/Fig20_wind_quiver_animation.gif`.

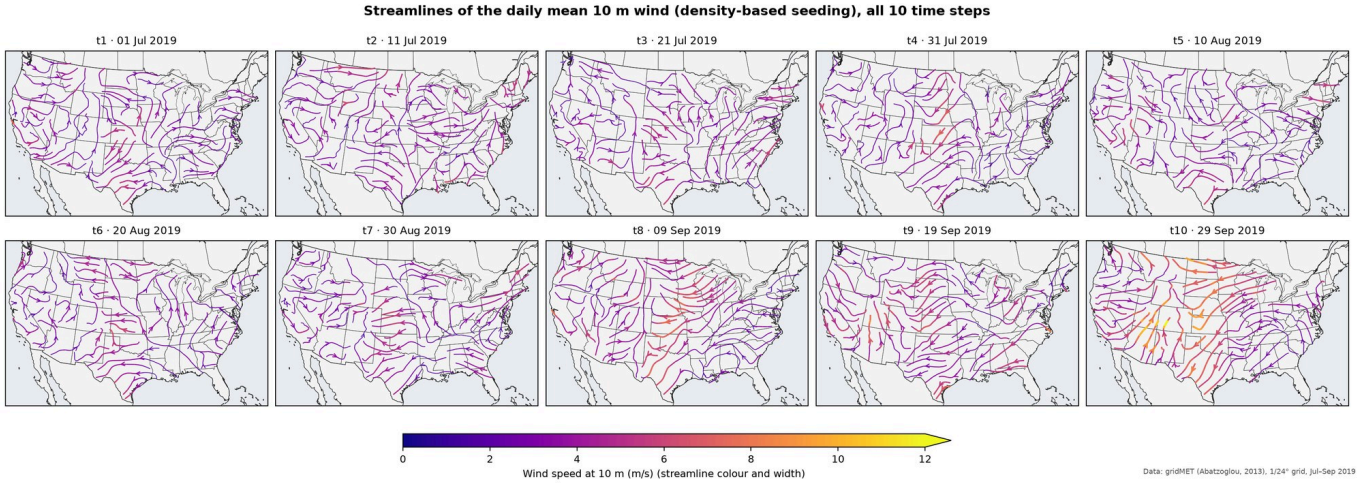


Fig. 21. Streamlines (density-based seeding; colour and width = speed) for all ten time steps. Animation: `images/Fig21_wind_streamlines_animation.gif`.

G. Storm close-ups between the samples

The ten-day sampling misses the three tropical cyclones of the quarter (Sec. II-B). Fig. 22 shows them on the 4 km grid on the day of strongest impact, with corrected quiver plots over wind speed (top) and streamlines over log-scaled precipitation (bottom). To quantify what the panels show we computed the relative vorticity and divergence of the wind after a 36 km mask-aware box smoothing.

- **Barry, 13 Jul** (landfall in Louisiana [8]). The strongest daily-mean winds of the three storms, up to 15.0 m/s, blow over the Mississippi delta. North-easterlies over Mississippi turn into easterlies across northern Louisiana, and onshore southerlies at the coast meet them at the 223 mm rain maximum (29.8° N, 92.1° W): 71 % of the area with ≥ 100 mm is convergent (mean divergence -7.7×10^{-5} s^{-1}). The closed circulation, however, lies over the Gulf, where gridMET has no data. Over land the flow curves clockwise as it weakens inland, so the land-mean vorticity is even slightly negative (-0.8×10^{-5} s^{-1}).
- **Dorian, 5 Sep** (centre offshore, moving up the coast). Easterlies of 9–10 m/s blow onshore over North Carolina, south-westerlies follow the South Carolina coast and north-westerlies blow offshore over Georgia. The streamlines wrap counter-clockwise around north-eastern South Carolina, next to the 251 mm maximum (33.5° N, 79.1° W), and the land-mean vorticity ($+0.7\times10^{-5}$ s^{-1}) is higher than on any of the ten sampled days in the same window (at most +0.6).
- **Imelda, 18 Sep** (slow-moving remnant over south-east Texas). The wind is weak, at most 7.7 m/s and 3.4 m/s on average, yet this is the wettest day of the quarter (707 mm at 29.9° N, 94.2° W; 14,656 km² with ≥ 100 mm). The streamlines show why: onshore southerlies from the Gulf converge with north-easterlies over eastern Texas over the rain maximum; 88 % of the ≥ 100 mm area is convergent (mean -6.7×10^{-5} s^{-1}, against $+0.5\times10^{-5}$ s^{-1} where it stayed dry).

Heavy rainfall is thus tied to sustained convergence and slow motion, not to wind speed. The close-ups also show the limits of the data: a daily-mean, land-only wind field smears a moving storm and cannot show a closed hurricane circulation.

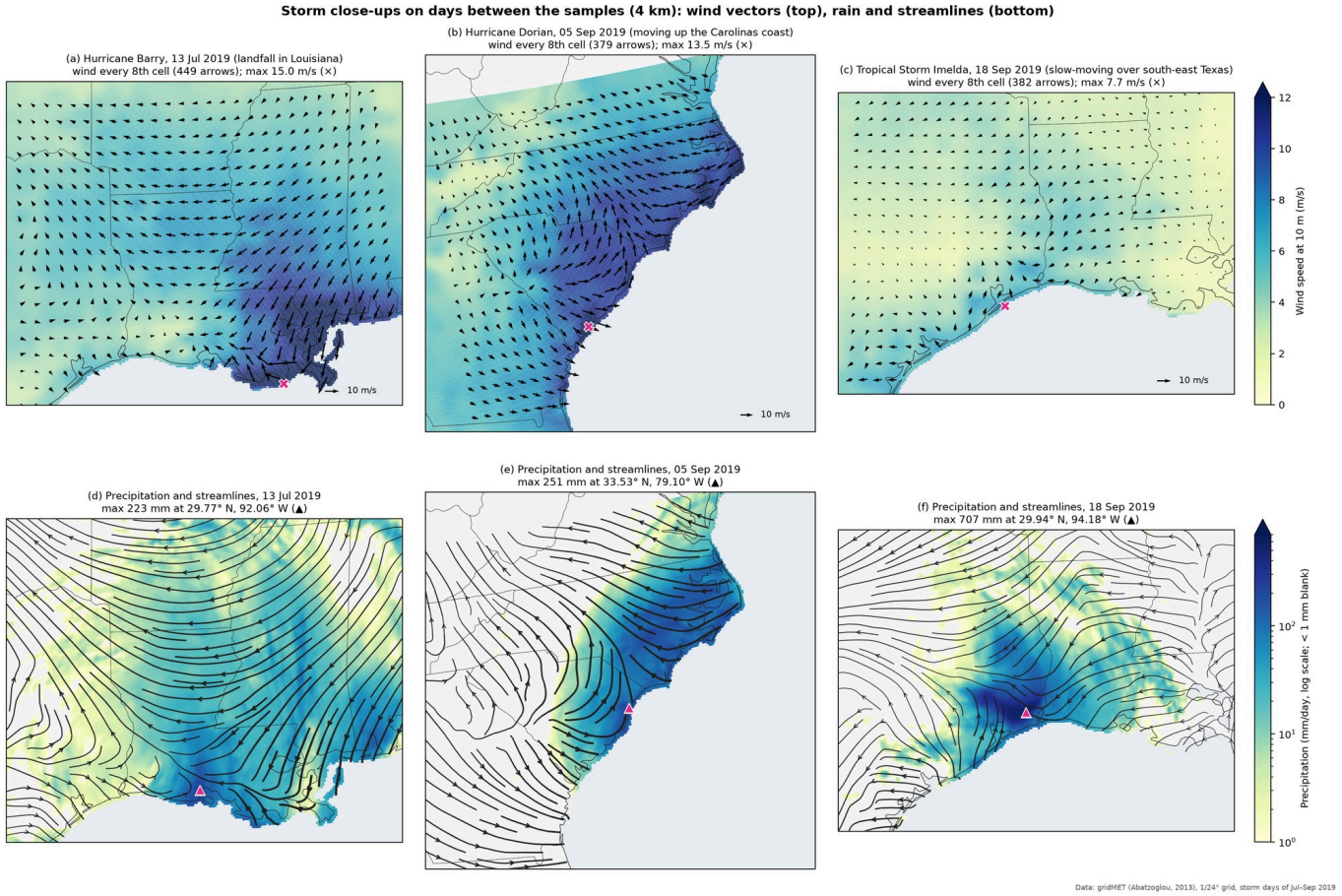


Fig. 22. Storm close-ups at 4 km on days between the samples: (a, d) Hurricane Barry, 13 Jul; (b, e) Hurricane Dorian, 5 Sep; (c, f) Tropical Storm Imelda, 18 Sep 2019. Top: wind speed with magnitude-proportional arrows every 8th cell (× = speed maximum). Bottom: precipitation (log scale) with streamlines (▲ = rainfall maximum).

VI. InfoVis Task 1: Node-Link Diagram *(Kalpit)*

A. Building the network

The network is built, analysed and laid out with NetworkX [26]. Each of the 167 countries is described by the share of its 1970–2022 events in 15 hazard categories: riverine, flash and coastal flood; tropical cyclone, extra-tropical storm and convective storm; heat wave; cold wave or winter; earthquake; volcanic activity; drought; landslide; avalanche; wildfire; and insect infestation. Unspecified floods and storms were re-assigned in proportion to each country's specified subtypes (Table III). The similarity of two profiles p, q is the Bhattacharyya coefficient [20]

$$\mathrm{BC}(p, q) = \Sigma_k \sqrt{(p_k\, q_k)} \in [0, 1], \qquad (2)$$

which compares distributions and is not dominated by the largest category. Each country is linked to its four most similar countries if BC ≥ 0.70, and the maximum spanning tree of the complete similarity graph is added so that every country is reachable. The result has 474 edges (472 nearest-neighbour edges and 2 spanning-tree edges), density 0.034, mean degree 5.7 (maximum 12, Germany and Niger), average clustering 0.37 and diameter 10, which makes it a sparse but well-connected network. Louvain community detection [21] on the similarity-weighted graph finds 8 communities with modularity Q = 0.69.

B. Layouts

Fig. 23 draws the network with the Fruchterman–Reingold force-directed layout [22], coloured by community (a) and by continent (b). The communities are hazard regimes, not geographic regions. C2 (tropical cyclones, 30 countries) groups Caribbean and Pacific islands with Japan, the Philippines, Taiwan, South Korea, Vietnam, Mexico, Madagascar and Mauritius. C4 (riverine flood and earthquake, 22 countries) follows the Alpine–Himalayan and Andean seismic belts from Italy, Turkey and Iran to Nepal, Indonesia, Peru and New Zealand and spans all five continents. C3 (riverine flood and drought) is almost entirely the Sahel and southern Africa (23 of 25). C5 (extra-tropical storm and riverine flood) is all of north-western and central Europe. Large, multi-hazard countries such as China, India, the US and Indonesia sit in the middle because their mixed profiles resemble many others; India's most similar countries are Bangladesh (0.95), China (0.94), Thailand (0.92), Pakistan (0.91) and Russia (0.90). The countries with the highest betweenness, Haiti (0.15), Romania, Bosnia and Herzegovina, Thailand and Niger, are bridges whose profiles mix two regimes. The labels of the most active countries are placed by a greedy collision-free algorithm: in order of events, each label tries eight positions around its node on three rings (with a leader line beyond the first) and is kept only where it overlaps neither another label nor a labelled node. Fig. 24 shows four more layouts: Kamada–Kawai [23], ForceAtlas2 [24], spectral [25] and a circular layout ordered by community.

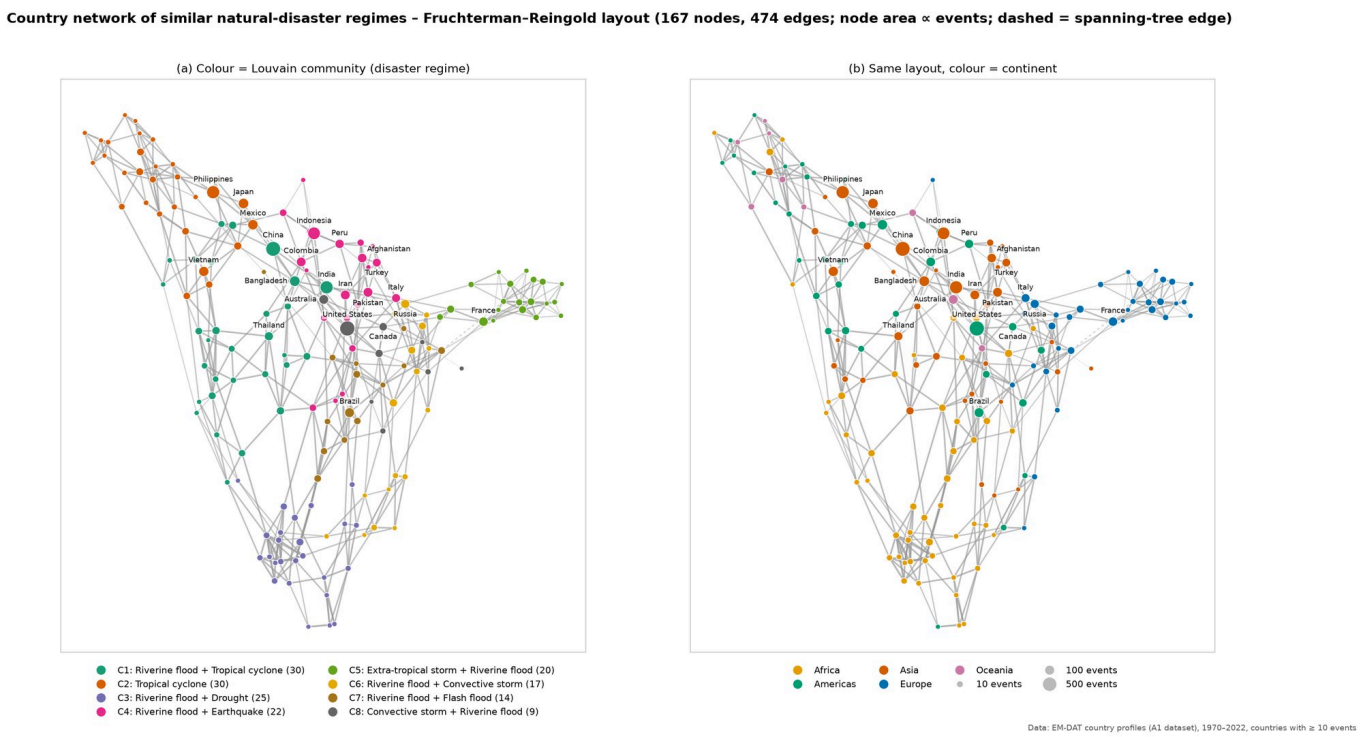


Fig. 23. Country network of similar natural-disaster regimes, Fruchterman–Reingold layout, coloured by (a) Louvain community and (b) continent; node area ∝ number of events; dashed edges belong to the maximum spanning tree.

C. Quantitative comparison

We measured each layout with six criteria, computed on positions normalised to the unit square (Fig. 25, Table VII). ForceAtlas2 with distributed attraction gives the tightest, most readable clusters, with the fewest crossings (754) and the best neighbourhood preservation (0.36): for each node, the overlap between its graph neighbours and its nearest neighbours in the drawing. Fruchterman–Reingold is close behind and has the most uniform edge lengths. Kamada–Kawai places nodes by graph-theoretic distance. It preserves global distances well (low stress) but lets communities overlap, giving the lowest silhouette (0.20) and the most crossings among the force methods (1,010), and it is the slowest because it works on all pairs. The spectral layout runs in 18 ms and reveals the network's main gradient as a "V": one arm is extra-tropical Europe, the vertex is the flood and seismic core, and the other arm is the tropical-cyclone islands. Many nodes overlap at the vertex, however (edge-length variation 0.90). The circular layout separates communities perfectly by construction (silhouette 0.45), but its inter-community edges become a hairball of 5,756 crossings, 6.7 times more than Fruchterman–Reingold. No layout wins on every criterion. Force-directed layouts are best for exploring clusters, spectral for the dominant ordering, and circular for comparing groups.

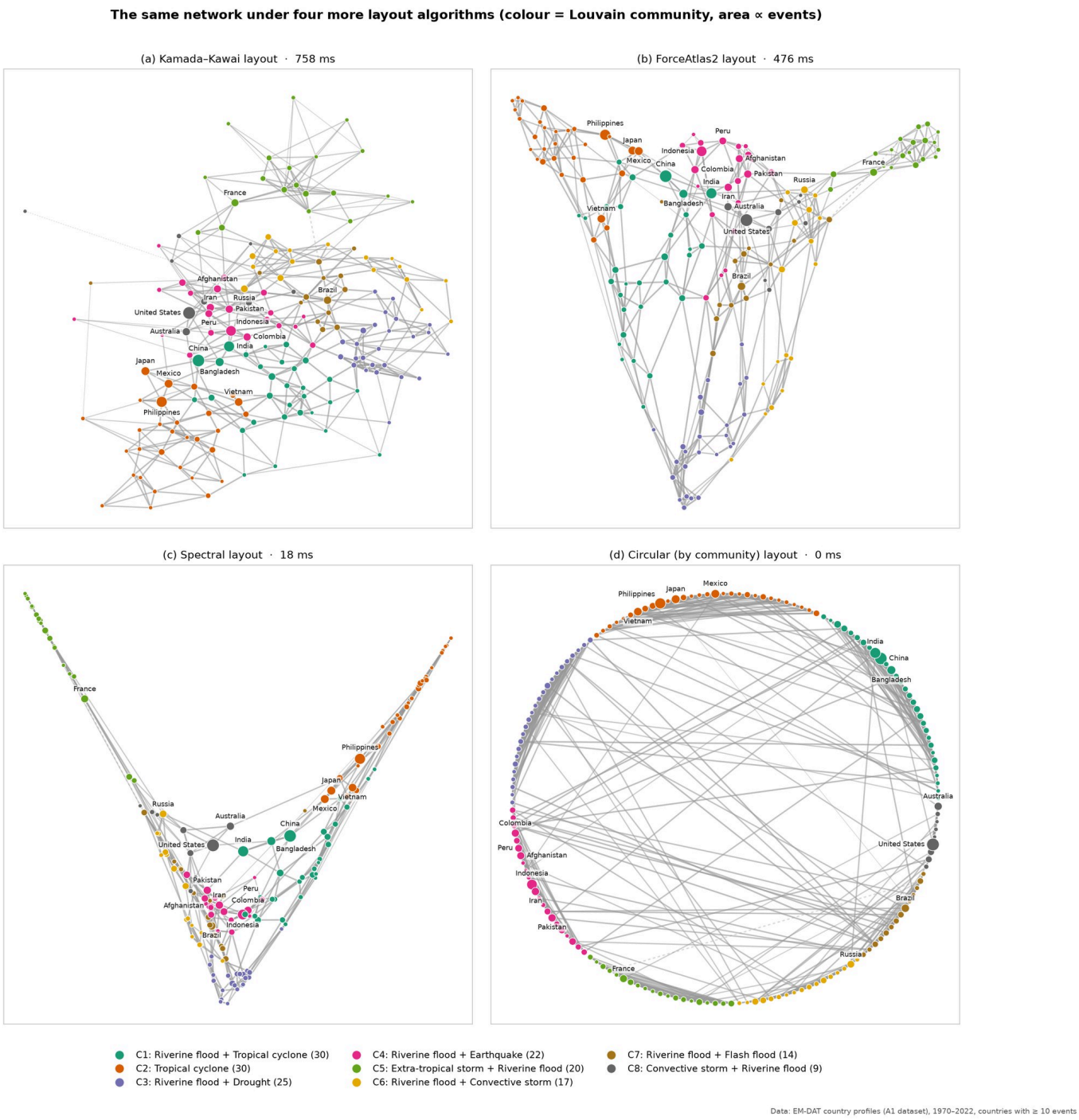


Fig. 24. The same network under (a) Kamada–Kawai, (b) ForceAtlas2, (c) spectral and (d) circular (ordered by community) layouts, coloured by community, with run times.

TABLE VII
LAYOUT QUALITY (167 NODES, 474 EDGES; ↓ LOWER IS BETTER, ↑ HIGHER IS BETTER)

Layout	*Run time (s)* ↓	*Edge crossings* ↓	*Normalised stress* ↓	*Edge-length CV* ↓	*Neighbourhood pres.* ↑	*Community silhouette* ↑
Fruchterman–Reingold	0.27	858	0.0009	0.50	0.31	0.26
Kamada–Kawai	0.76	1,010	0.0010	0.67	0.33	0.20
ForceAtlas2	0.48	754	0.0014	0.66	0.36	0.27
Spectral	0.02	797	0.0023	0.90	0.29	0.29
Circular (by community)	< 0.01	5,756	0.0101	1.10	0.17	0.45

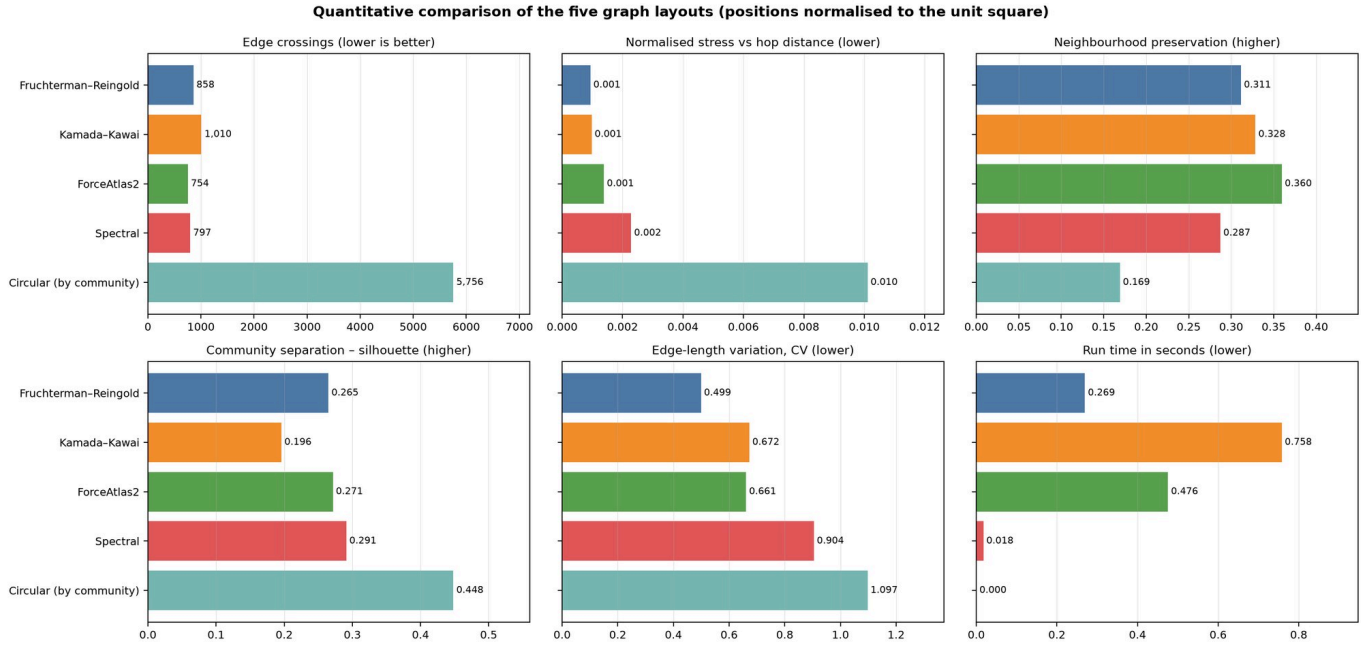


Fig. 25. Quantitative comparison of the five layouts.

D. What the communities mean

Fig. 26 confirms the interpretation. The tropical-cyclone community draws on average 66 % of its events from tropical cyclones, and the Sahel community 58 % from riverine floods, most of the rest being droughts. The European community is 100 % European and the Sahel community 92 % African, whereas the cyclone, seismic and convective-storm communities each span three or more continents. Disaster regimes follow climate and plate tectonics, not political geography.

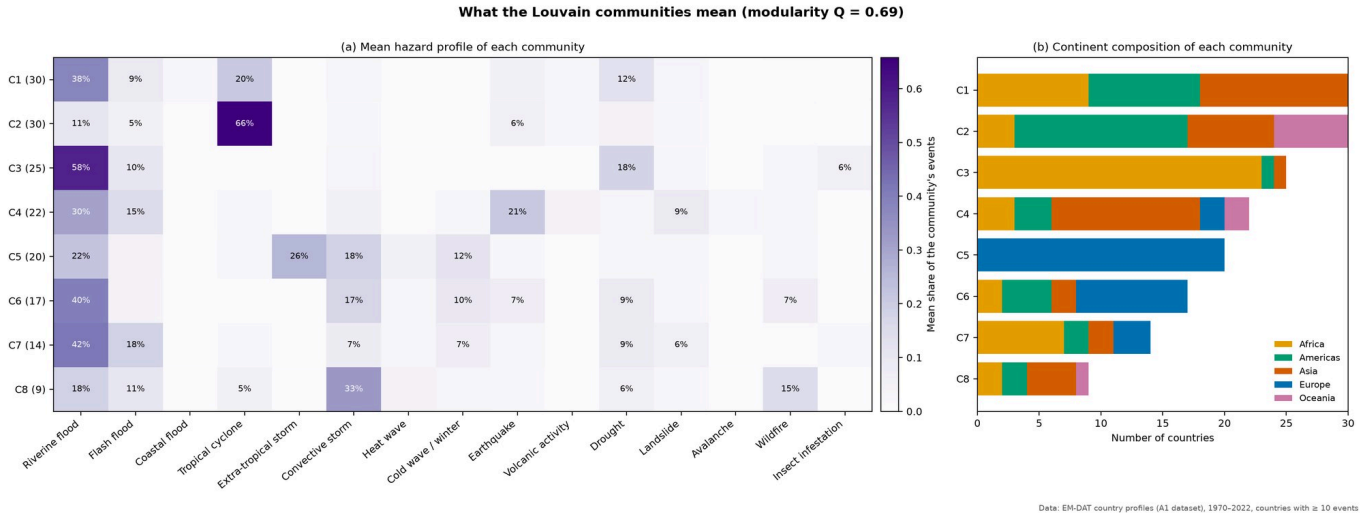


Fig. 26. (a) Mean hazard profile and (b) continent composition of the eight Louvain communities.

E. User interaction

The interactive viewer (`code/web/network.html`) uses the positions computed by NetworkX and adds user interactions (Fig. 27). The user can switch between the five layouts with animated transitions, or run a live d3-force simulation. Nodes can be coloured by community or continent and sized by events, deaths, degree or betweenness. Edges can be filtered by minimum similarity and nodes by minimum number of events, and continents can be toggled on and off. There is a search box, clicking a node focuses on its neighbourhood (with its hazard profile and most similar countries listed), and the view supports zooming, panning and dragging nodes. Filtering is especially useful for node-link diagrams. Keeping only edges with BC $\geq$ 0.9 and countries with $\geq$ 50 events (Fig. 27c) reduces the drawing to a backbone of 73 countries and 123 edges, in which the European group (top), the tropical-cyclone group (bottom left) and the flood–seismic core separate clearly. Labels in the viewer are placed with the same collision-avoiding strategy. The network is also exported in GEXF format (`data/processed/network_disaster_regimes.gexf`) with all node and edge attributes, the ForceAtlas2 positions, community colours and event-based sizes, so that it opens ready-styled in Gephi [39] for further editing.

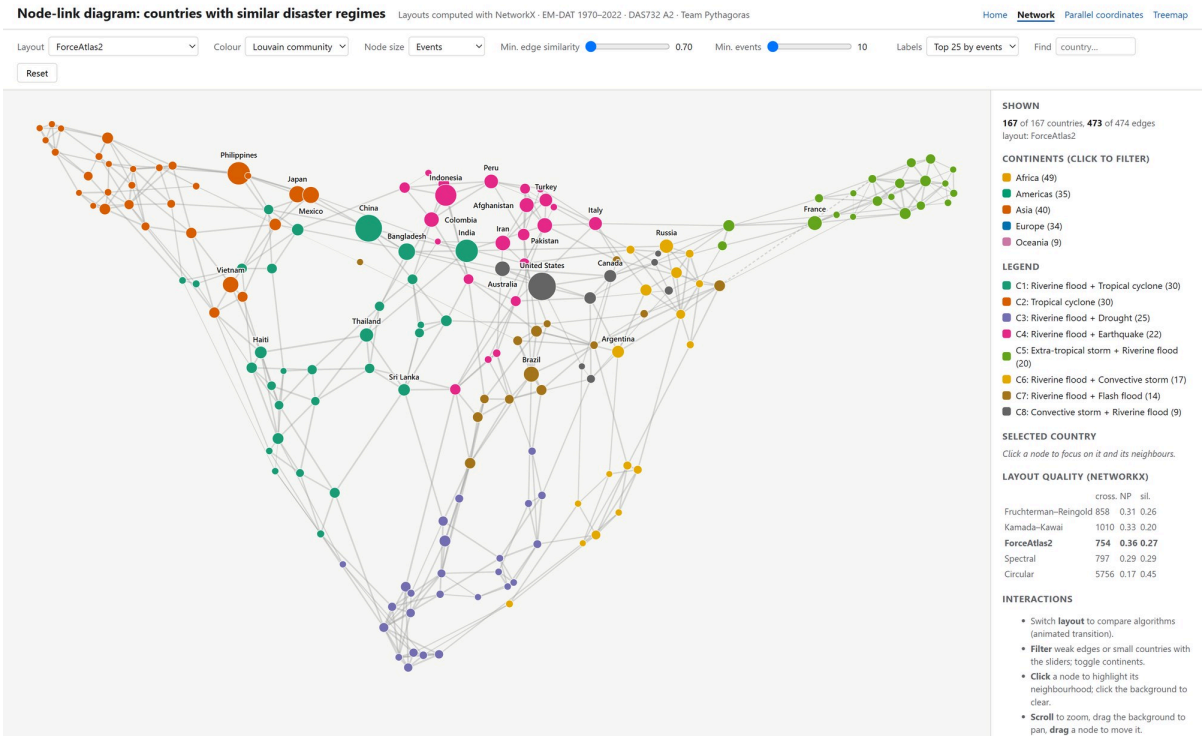


Fig. 27(a). Interactive network viewer: ForceAtlas2 layout coloured by community, with the legend and the layout-quality panel.

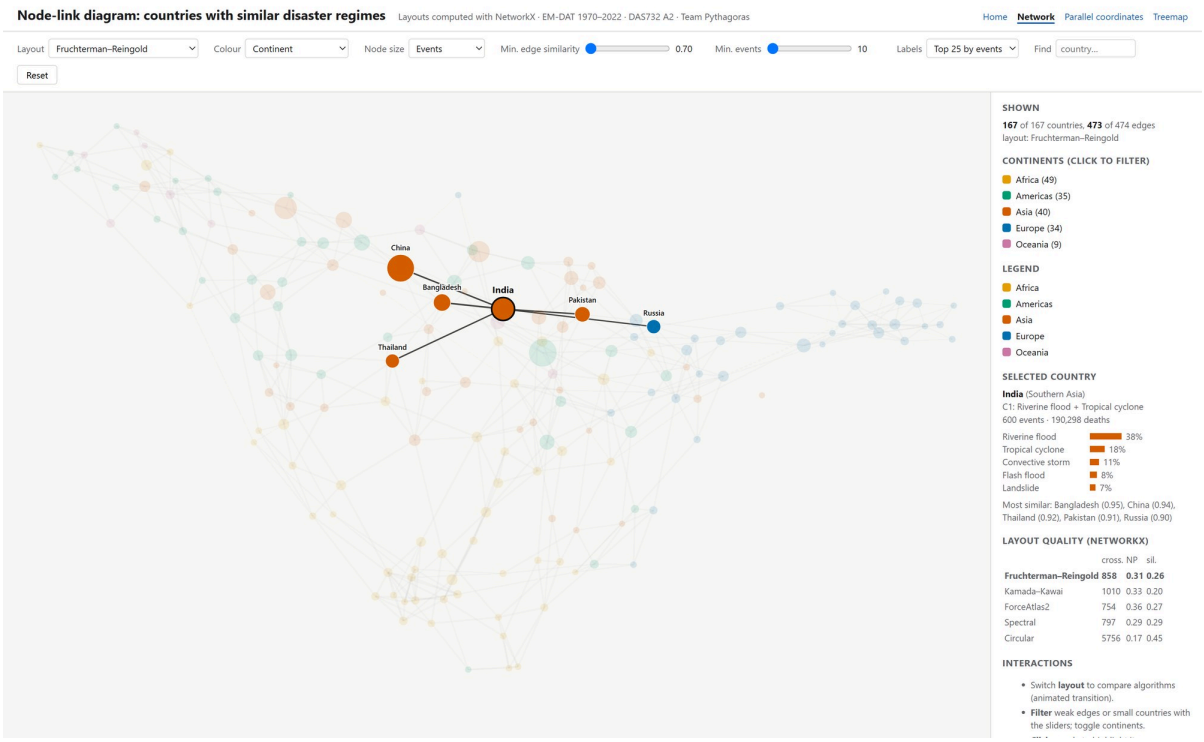


Fig. 27(b). Focus + context: clicking India highlights its five most similar countries and shows its hazard profile (Fruchterman–Reingold layout, continent colouring).

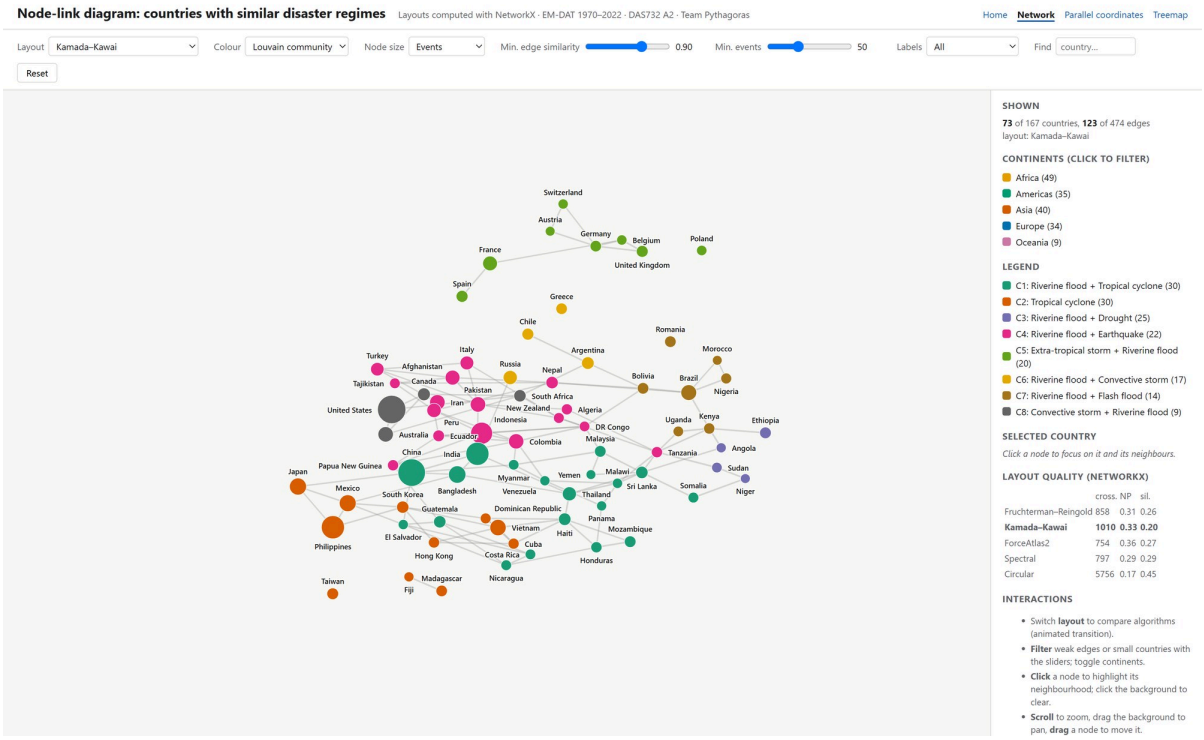


Fig. 27(c). Filtering: edges with similarity ≥ 0.90 and countries with ≥ 50 events only, Kamada–Kawai layout with all labels.

VII. InfoVis Task 2: Parallel Coordinates and Zoomable Treemap *(Lakshya Jain)*

A. Multivariate data and clusters

The parallel-coordinates plot (PCP) [28] (`code/web/parallel_coordinates.html`, implemented in D3 [27], which is bundled in `code/web/lib/` so that all pages work offline) shows the 12 dimensions of Table IV for the 167 countries, or, in a second view, for 622 country-decades (1970s–2010s, ≥ 3 events), where growth is replaced by a decade axis. Heavy-tailed variables use symmetric-log axes so that zeros stay visible. Rows were clustered with k-means (k-means++ initialisation, 25 restarts) on standardised features, with log-transformed counts. k was chosen by silhouette. The country view uses $k = 3$ (silhouette 0.24, against 0.19–0.22 for $k = 4$–7). The decade view uses $k = 5$, which we preferred because its silhouette lies within 0.02 of the best. The three country clusters are:

- flood-dominated, under-reported countries (68; median hydrological share 63 %, damage reported for only 18 % of records, 36 of them African);
- high-impact, high-volume countries (52; median 108 events, 20,349 deaths, 114 deaths per event and seven hazard types, e.g. the US, China, the Philippines, India, Indonesia and Bangladesh);
- storm-dominated countries (47; median meteorological share 67 %, six deaths per event and the highest damage reporting, 45 %).

B. Edge bundling and interactions

We implemented cluster-based curved bundling following the illustrative and edge-bundling PCP literature [29]–[31]. For a row in group g, between adjacent axes i and i+1, the polyline passes through a midpoint

$$y_{\mathrm{m}} = (1 - \beta)\,(y_i + y_{i+1})/2 + \beta\,(c_{g,i} + c_{g,i+1})/2, \qquad (3)$$

where $c_{g,i}$ is the group's centroid on axis i and $\beta \in [0, 1]$ is the bundling strength. The axis points and midpoints are joined by cubic Bézier segments whose control points are offset by a smoothness parameter s. With $\beta = s = 0$ the plot reduces exactly to classical straight polylines. Bundling can be by k-means cluster or by continent. The plot supports all the required interactions:

- **brushing** on any number of axes, combined with AND, with live counts and a list of the selected rows;
- **axis reordering** by dragging an axis title, and **axis inversion** by double-clicking it;
- the β and s sliders, legend filtering, search and hover details.

The view state can be set in the URL, so every screenshot here can be reproduced.

C. Insights from the PCP

Without bundling (Fig. 28), 167 lines overplot heavily. The four subgroup-share axes sum to about one, which produces X-shaped crossings (hydrological vs. meteorological share: $r = -0.86$), and the clusters are hard to follow beyond the first axes. With $\beta = 0.8$ (Fig. 29) the three clusters become three bundles whose trajectories can be read end to end. The orange high-impact bundle runs along the top of the four impact axes and the hazard-type axis. The green flood bundle rises on hydrological share and falls on damage reporting. The purple storm bundle rises on meteorological share and on damage reporting. The price is accuracy: values are exact only on the axes, never between them.

Brushing turns the plot into a query tool. Selecting countries where at least 60 % of records report damage (Fig. 30a) returns 15 countries: the US, China, the Philippines, Vietnam, Australia, South Korea, Taiwan, Germany, Fiji, Jamaica, Tonga, the Bahamas, Belize, Sweden and Samoa.

Nearly all are storm-prone, high-income, or both. Across all countries, damage reporting correlates with meteorological share (r = 0.62) and anti-correlates with hydrological share (−0.52), and its median is 13 % in Africa against 40 % in Europe and 55 % in Oceania. The damage totals therefore over-represent cyclone-exposed and rich countries [37]. Africa's 1.3 % share of 1970–2022 damage is partly a reporting artefact and should not be read as low economic loss. Brushing deaths per event ≥ 100 (Fig. 30b) selects 29 countries (13 Asian, 7 American, 5 European, 4 African). At the top are Ethiopia (4,425 deaths per event, from drought famines), Sudan, Myanmar (Cyclone Nargis), Haiti (the 2010 earthquake), Bangladesh and Mozambique. The European entries (France, Italy, Russia and Spain) come from heat-wave mortality, notably in 2003 and 2010. The decade view (Fig. 31) has its axes reordered so that decade comes first, next to the shares, and its damage-reporting axis inverted. It shows the median deaths per event falling from 29.7 in the 1970s to 10.9 in the 2010s. Over the same decades the flood-dominated cluster grows from 26 to 59 country-decades, while the deadly geophysical cluster shrinks from 17 to 7. The growth axis also contains a data artefact. Its five highest values (Czechia 33, Croatia 32, Bosnia and Herzegovina 29, Serbia 28 and Slovakia 24, against a median of 2.7) belong to states created in 1991–93. Their earlier events are recorded under Yugoslavia or Czechoslovakia, so the growth is state succession, not hazard.

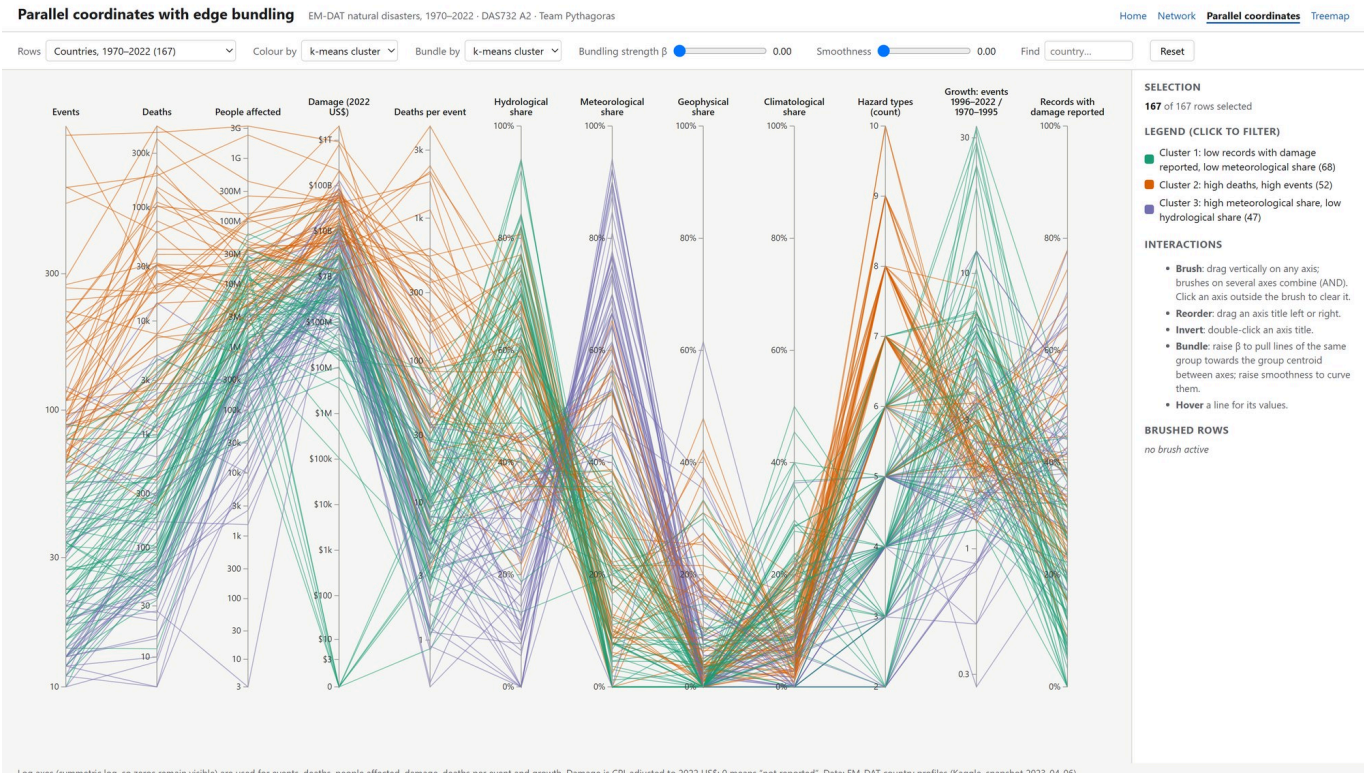


Fig. 28. Parallel coordinates of 167 countries (1970–2022), coloured by k-means cluster, without bundling (β = 0).

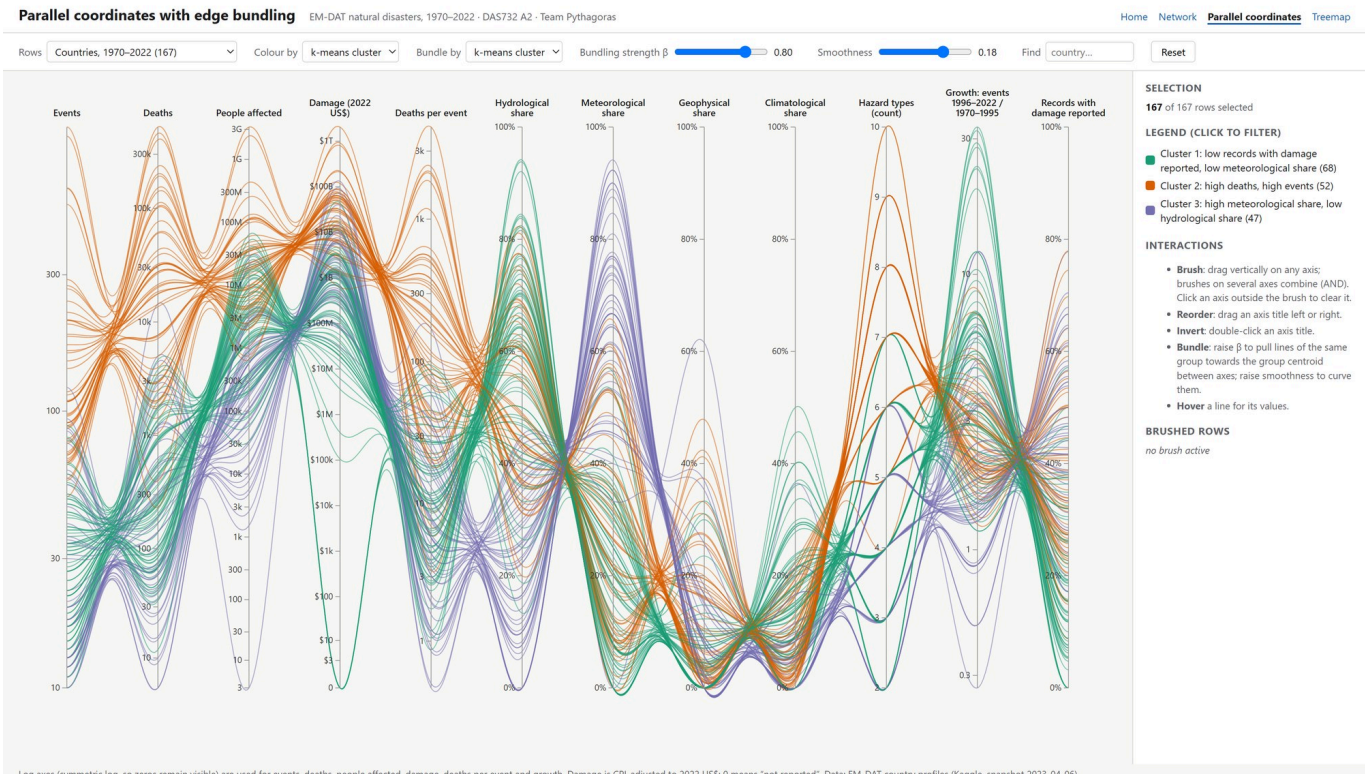


Fig. 29. The same plot with cluster-based edge bundling (β = 0.8, smoothness 0.18).

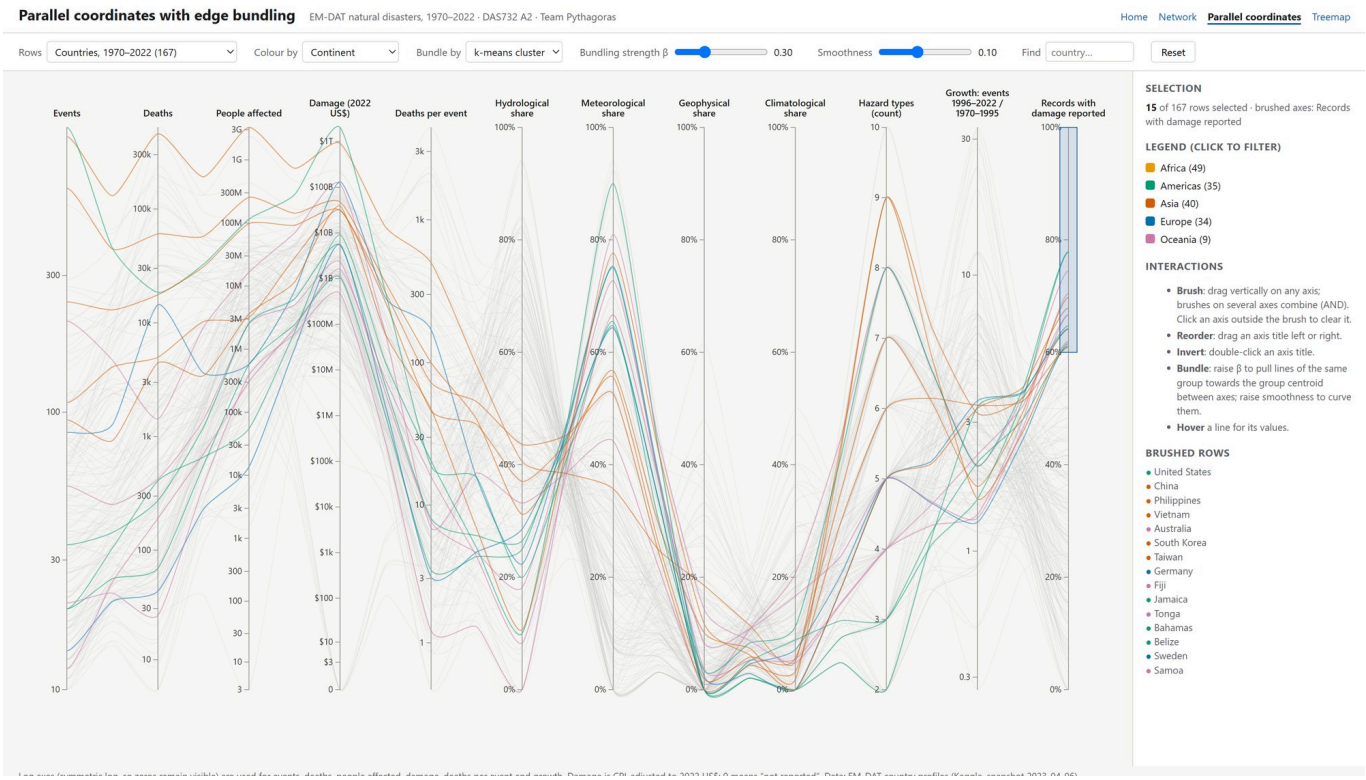


Fig. 30(a). Brushing: countries with damage reported for ≥ 60 % of records (15 of 167), coloured by continent; the side panel lists the selection.

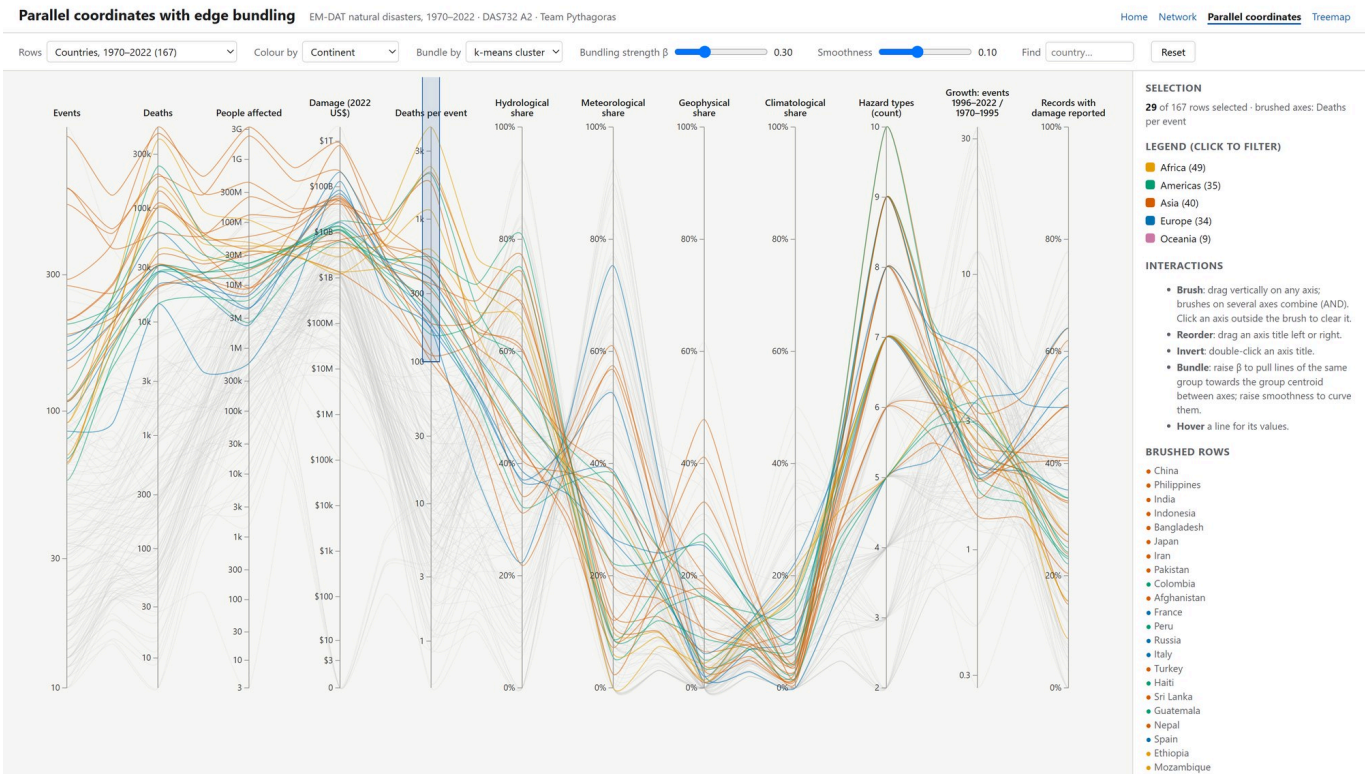


Fig. 30(b). Brushing: countries with ≥ 100 deaths per event (29 of 167).

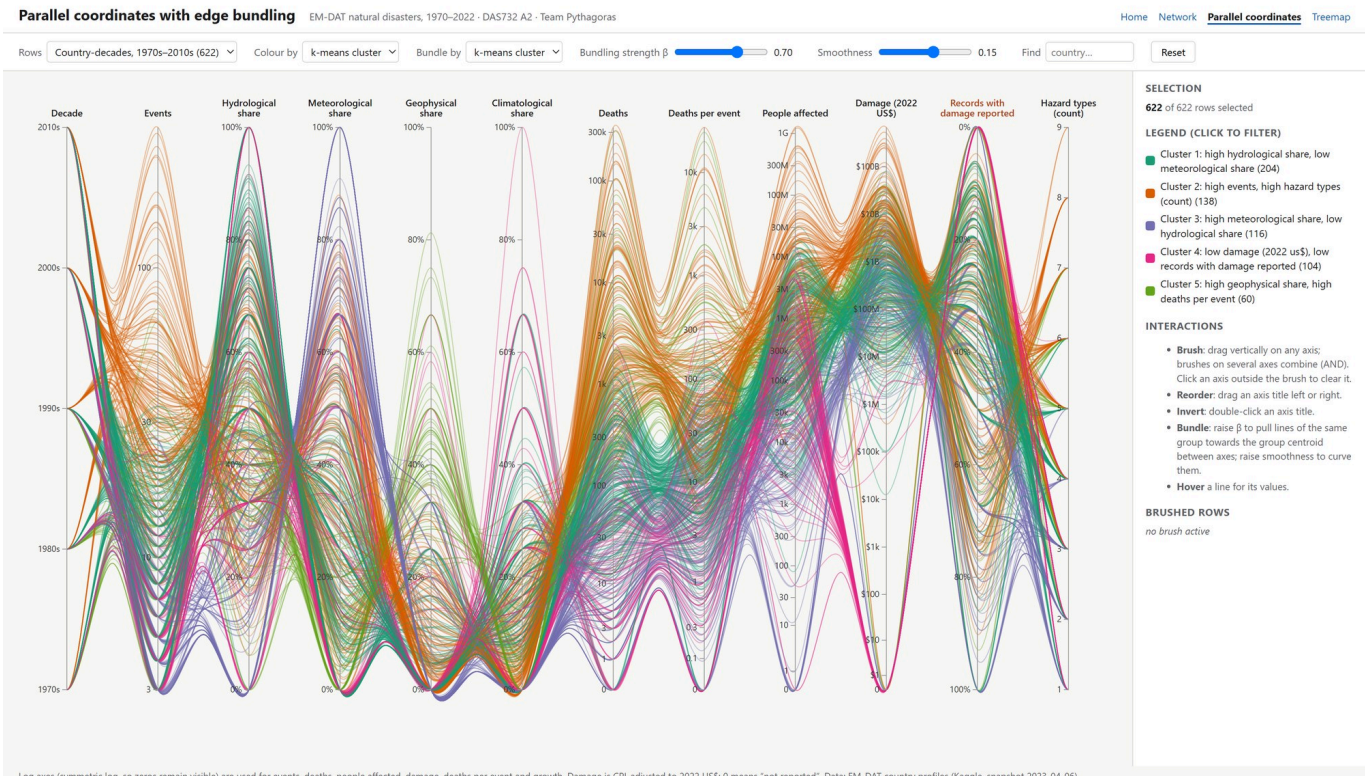


Fig. 31. Country-decade view (622 rows) with reordered axes (decade first, shares next), inverted damage-reporting axis and bundling by the five k-means clusters (β = 0.7).

D. Zoomable treemap: remodelling the table into trees

A treemap [32], [33] needs a tree, while the A1 data is a flat table. Our treemap (`code/web/treemap.html`) builds the tree in the browser from the 4,937-row rollup using a user-selected order of categorical columns, which gives four remodels of the same table:

- Continent › Sub-region › Country › Hazard type;
- Hazard group › Type › Subtype › Continent;
- Hazard group › Type › Continent › Country;
- Decade › Hazard group › Type › Subtype.

Leaf size can be events, deaths, people affected or damage, and a period filter (1900, 1970 or 2000 onwards) is available. Two levels are drawn at a time. Clicking a block zooms into it with an animated transition, breadcrumbs lead back up, and

hovering shows the value, the share of the parent and the share of the total.

E. Insights from the treemap

By events (Fig. 32, 1900–2022), Asia leads with 6,261 events (42 %), ahead of the Americas (26 %), Africa (14 %), Europe (13 %) and Oceania (4 %). Southern, South-eastern and Eastern Asia are almost equal (1,932, 1,909 and 1,811 events). Sizing by deaths and zooming into Asia › Southern Asia (Fig. 33) tells a different story. The sub-region holds 7.61 M deaths, 33 % of all disaster deaths since 1900. India's 4.59 M deaths are 93 % drought, dominated by three famine-droughts (1900, 1942 and 1965, with 1.25–1.5 M deaths each). Bangladesh's 2.59 M deaths split into drought (1.90 M, the 1943 Bengal famine) and storms (0.63 M, the 1970 and 1991 cyclones). A few historical famines thus dominate the century-long toll. Restricted to 1970–2022, the ranking changes to Bangladesh (0.52 M), China, Ethiopia, Haiti, Indonesia and India (0.19 M).

Re-sizing the hazard tree by measure (Fig. 34, 1970–2022) answers "which hazard matters most?" differently each time. Hydrological hazards are the most frequent (46 % of events), and floods affect the most people (47 %). Geophysical hazards cause the most deaths (39 %; earthquakes alone 38 %) from only 9 % of events. Meteorological hazards cause the most damage (44 %; storms 42 %). The time hierarchy (Fig. 35) shows recorded events growing from 74 in the 1900s to 808 in the 1970s and 3,872 in the 2000s (the peak), with 3,504 in the 2010s. Geophysical events make up 66 % of the 1900s records but only 9 % of the 2010s records, which confirms that the early record favours large, well-documented earthquakes and eruptions.

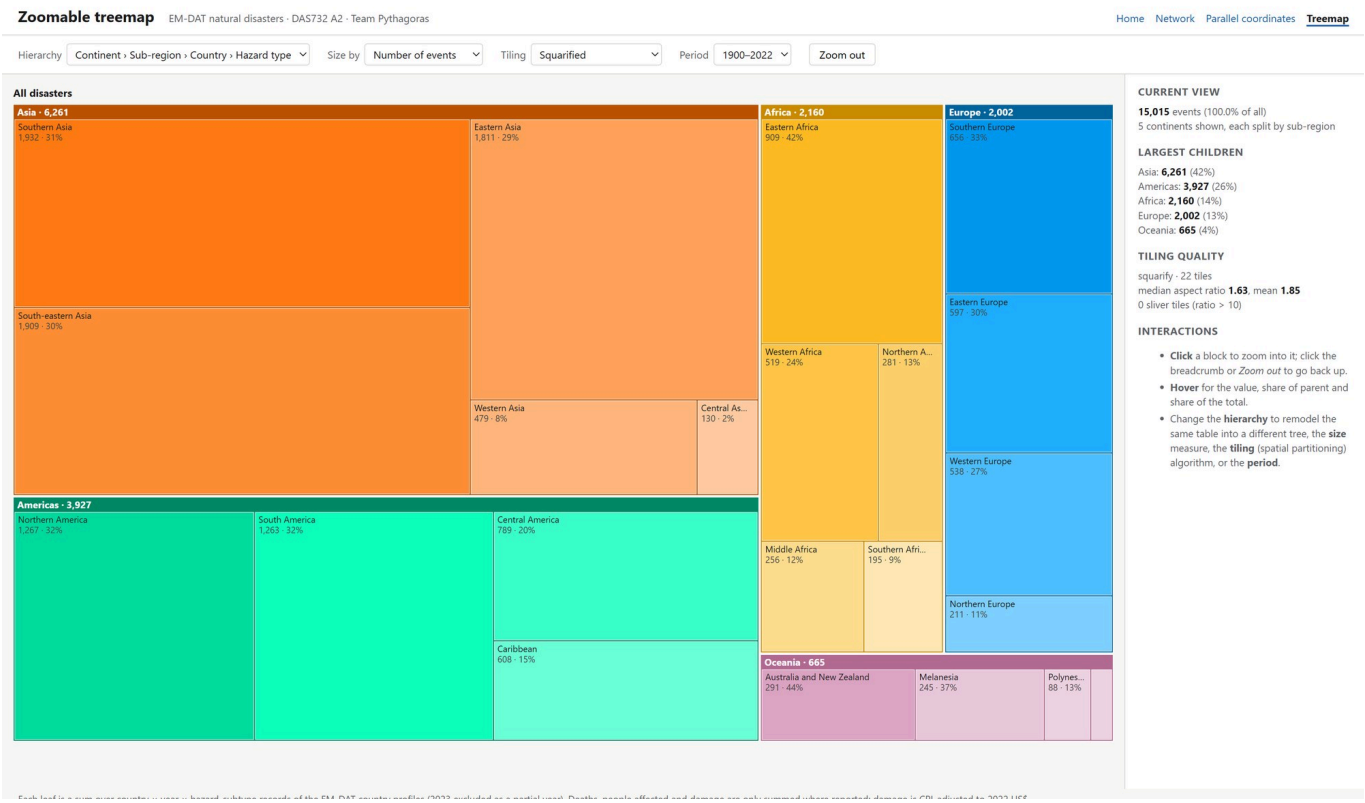


Fig. 32. Zoomable treemap, hierarchy Continent › Sub-region › Country › Type, sized by events (1900–2022, squarified tiling).

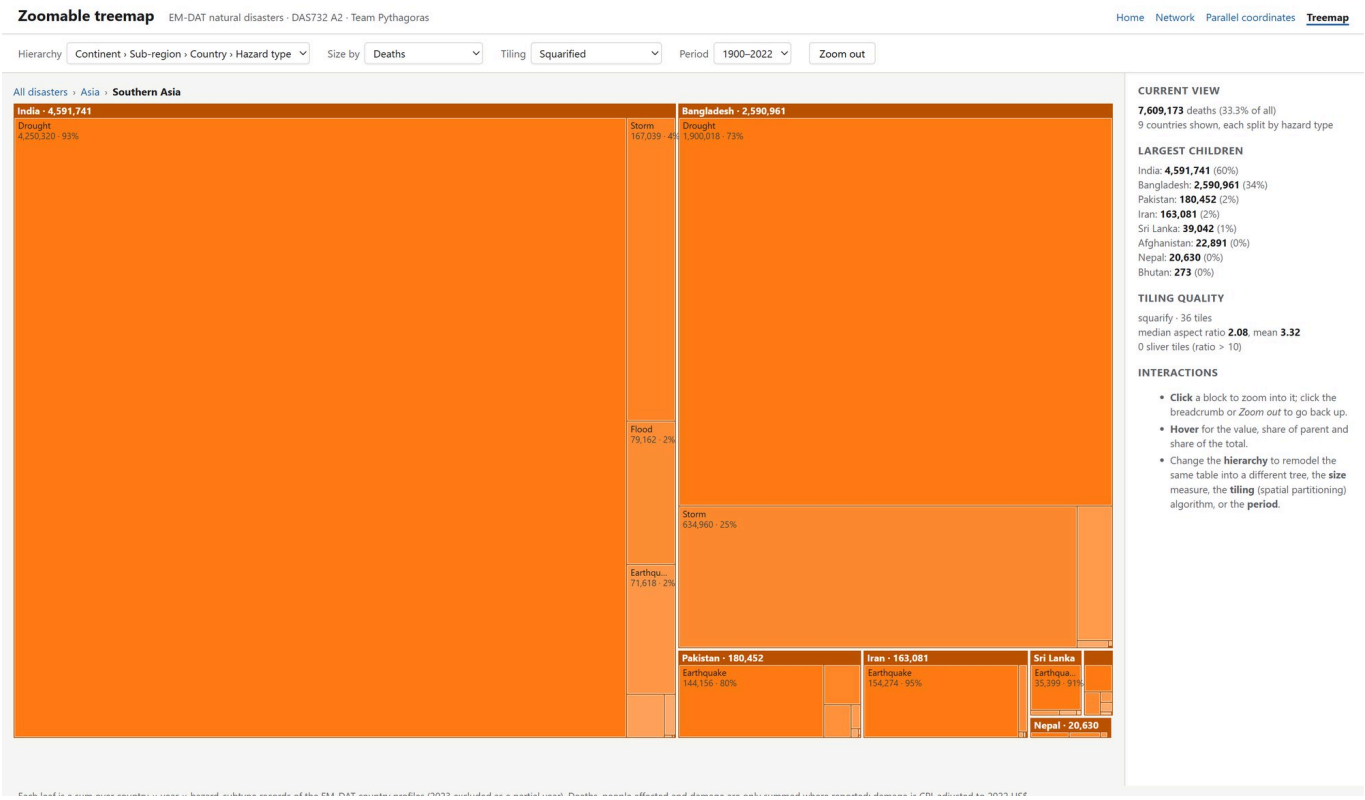


Fig. 33. After zooming into Asia › Southern Asia, sized by deaths (1900–2022): countries split by hazard type.

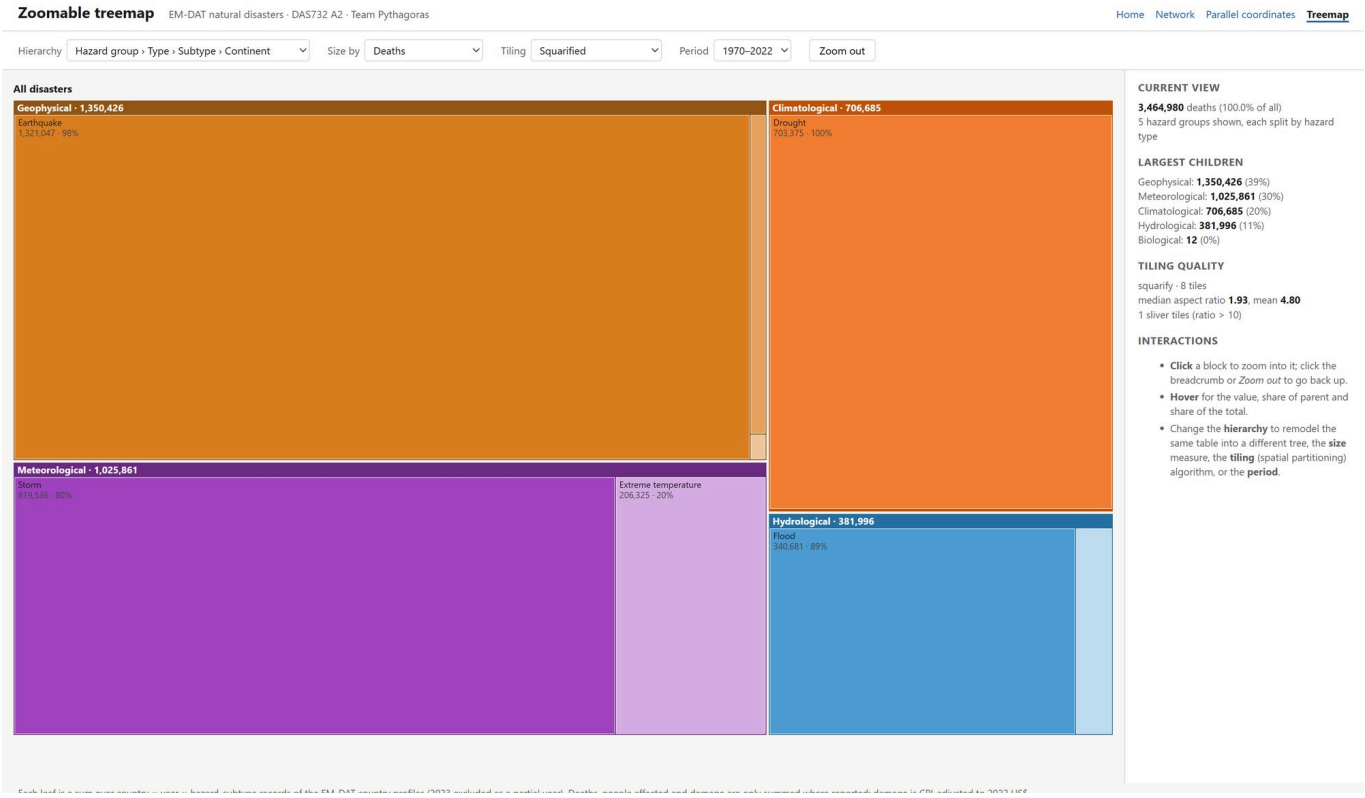


Fig. 34(a). Hazard hierarchy (Group › Type › Subtype › Continent) sized by deaths, 1970–2022.

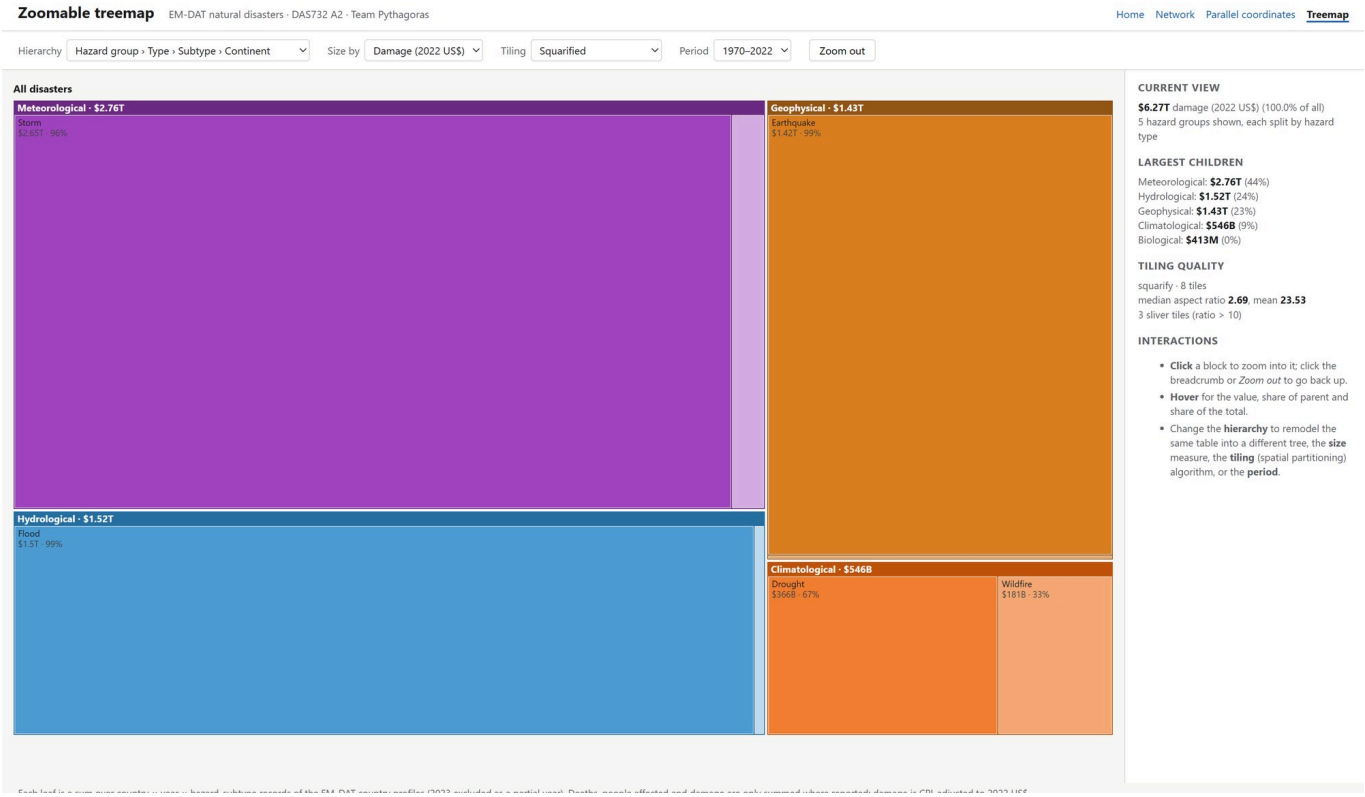


Fig. 34(b). The same tree sized by damage (2022 US$), 1970–2022.

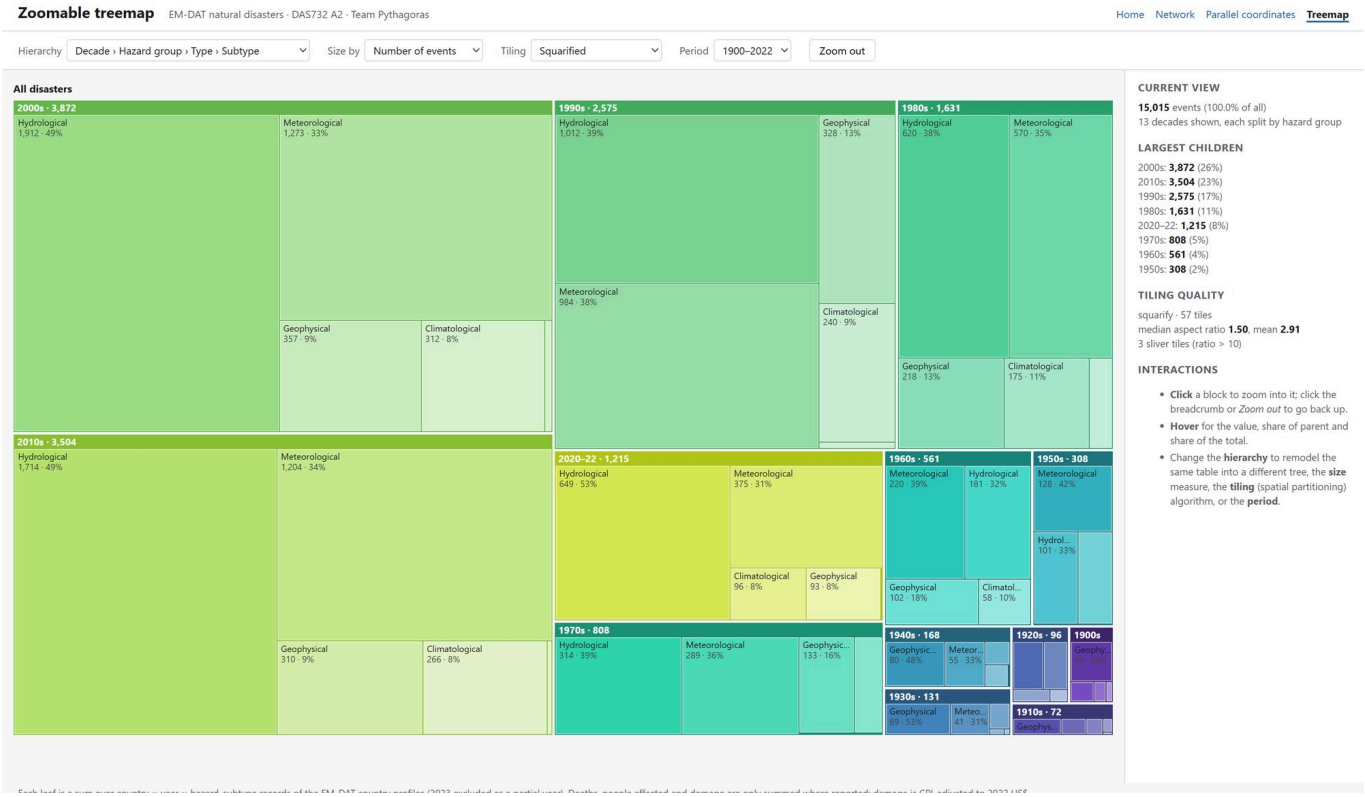


Fig. 35. Time hierarchy (Decade › Group › Type › Subtype) sized by events, 1900–2022.

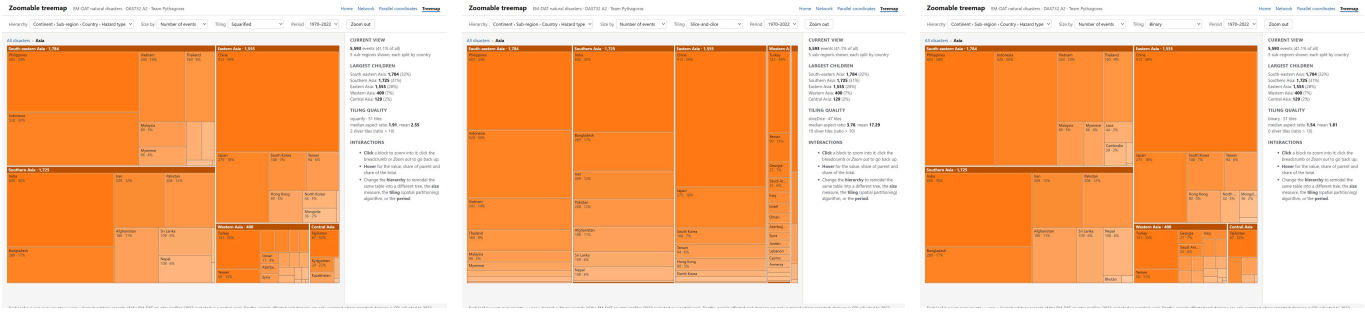
F. Spatial partitioning strategies

The treemap offers six tiling algorithms: squarified, binary, slice-and-dice, slice, dice and resquarified. The side panel reports the aspect ratios of the visible tiles. Fig. 36 and Table VIII compare three of them on Asia's sub-regions and countries (1970–2022). Slice-and-dice alternates the split direction by level, which preserves order and makes parent groups easy to see, but small countries become slivers. Ten tiles have an aspect ratio above 10, the mean ratio is 17.3, and four countries are too thin to render at all. Squarified tiling (target ratio $\varphi = 1.618$) produces compact tiles with a median ratio of 1.91 but scatters the order. Binary tiling, which splits the list into halves of equal weight, gave the most square tiles here (median 1.54, no slivers) while keeping a rough order. Resquarified tiling keeps tile positions

stable when the measure changes, which matters during interaction more than raw aspect ratio.

TABLE VIII
TILING QUALITY, ASIA › SUB-REGION › COUNTRY (EVENTS, 1970–2022)

Tiling	*Tiles*	*Median AR*	*Mean AR*	*AR > 10*
Squarified	51	1.91	2.55	2
Slice-and-dice	47	3.76	17.29	10
Binary	51	1.54	1.81	0



(a) Squarified (b) Slice-and-dice (c) Binary

Fig. 36. Spatial partitioning strategies on the same subtree (Asia › sub-region › country, events, 1970–2022).

VIII. InfoVis Task 3: Horizon and Ridgeline Plots *(Arnav)*

A. Methods

A horizon chart [34], [35] folds a time series into $n = 3$ bands of height $h = v_{\max}/3$. Band k shows clip($v - k\,h$, 0, h) in a darker shade, and negative values are mirrored upwards in red. Each series therefore occupies a third of the height of an area chart with the same resolution, so ten series fit in one figure. Ridgeline plots [36] stack kernel density estimates (Gaussian kernel, Scott bandwidth [38], weighted by the number of events) with partial overlap, each row normalised to the same peak height and sorted by its weighted median. As the source is yearly, the required variations are different groups (disaster types, continents, decades) and different temporal granularity (yearly vs. 5-year bins and decades). The year 2023 is excluded.

B. Horizon plots

Fig. 37 shows annual events per type, 1950–2022, each row scaled to its own maximum. Every type grows, but not equally. Before 1986, weather-related types had recorded only a small part of their 1950–2022 events: insect infestations 6.5 %, extreme temperature 8.3 %, floods 13.5 % and storms 20.2 %. For earthquakes and volcanic activity the share is 28 % and 27 %. Large geophysical events were always reported, so the much steeper growth of weather-related records since the mid-1980s reflects improved registration (EM-DAT was established in 1988) and rising exposure, as well as any change in hazard frequency. The dark bands mark peak years: floods in 2006 (226 events), extreme temperature in 2012 (52), drought in 2015 (27, the 2015–16 El Niño) and insect infestations in 1986–89 (the desert-locust upsurge).

Because the total number of records grows more than fourfold, from 81 events per year in the 1970s to 350 in the 2010s, raw counts make every row of Fig. 37 say “more disasters lately”. Fig. 38 removes this common factor. Each row shows the type's share of all events in a centred 5-year window minus its 1970–2022 share, on a common scale with bands of 2.5 percentage points (pp), blue above and red (mirrored) below. The mix of hazards changes. Floods rise from 32.1 % of the events in the 1970s to 43.7 % in the 2010s (more than 5 pp below their mean until the early 1990s, well above it since 2004), extreme temperature from 1.9 % to 5.9 % and wildfires from 0.9 % to 2.8 %. Earthquakes fall from 13.6 % to 7.5 %, storms from 33.9 % to 28.5 % and droughts from 7.5 % to 4.8 %. Shares are not free of reporting effects either (a type whose small events became easier to register gains share), but they remove the overall growth and show relative change, which raw counts cannot.

Fig. 39 changes both the group (continents) and the quantity: each year's deaths are shown as $\log_{10}$(deaths + 1) minus the continent's median year, in bands of 0.5 log units (≈ 3×), blue above and red below the median. Median annual deaths differ by more than two orders of magnitude, from 12,971 in Asia to 57 in Oceania. The two most extreme years of each continent are major catastrophes:

- Africa: 1983 (450,529 deaths, 475× its median; the Ethiopian and Sudanese drought-famine) and 1973;
- the Americas: 2010 (225,727; the Haiti earthquake) and 1970 (67,784; the Ancash earthquake in Peru);
- Asia: 1970 (317,526; the Bhola cyclone) and 1976 (254,994; the Tangshan earthquake);
- Europe: 2003 (72,298; 129×) and 2010 (56,472), both heat waves;
- Oceania: 1998 (2,215; the Papua New Guinea tsunami).

Asia's extremes are only about 20× its median because its baseline is high. Europe and the Americas, in contrast, are quiet with rare catastrophes. Since about 2011 Asia's row is mostly red, i.e. below its median, which is consistent with falling mortality from cyclones and floods.

Fig. 40 aggregates the data of Fig. 37 into 5-year bins (mean events per year, so the three-year bin 2020–22 is comparable). Year-to-year noise such as El Niño years disappears and the trend remains. Floods and storms peak in the most recent bin; earthquakes, landslides, droughts and wildfires peak in 2000–04, extreme temperature in 2010–14 and insect infestations in 1985–89. The yearly chart is better

for locating events, and the 5-year chart for comparing trends.

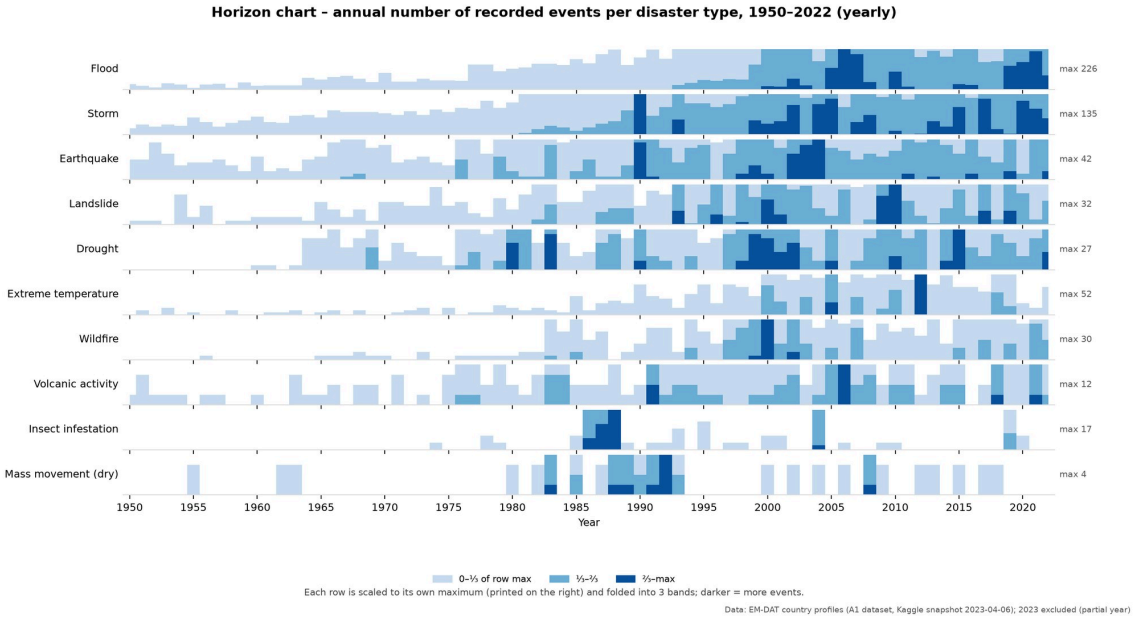


Fig. 37. Horizon chart of the annual number of events per disaster type, 1950–2022 (yearly; each row scaled to its own maximum, three bands).

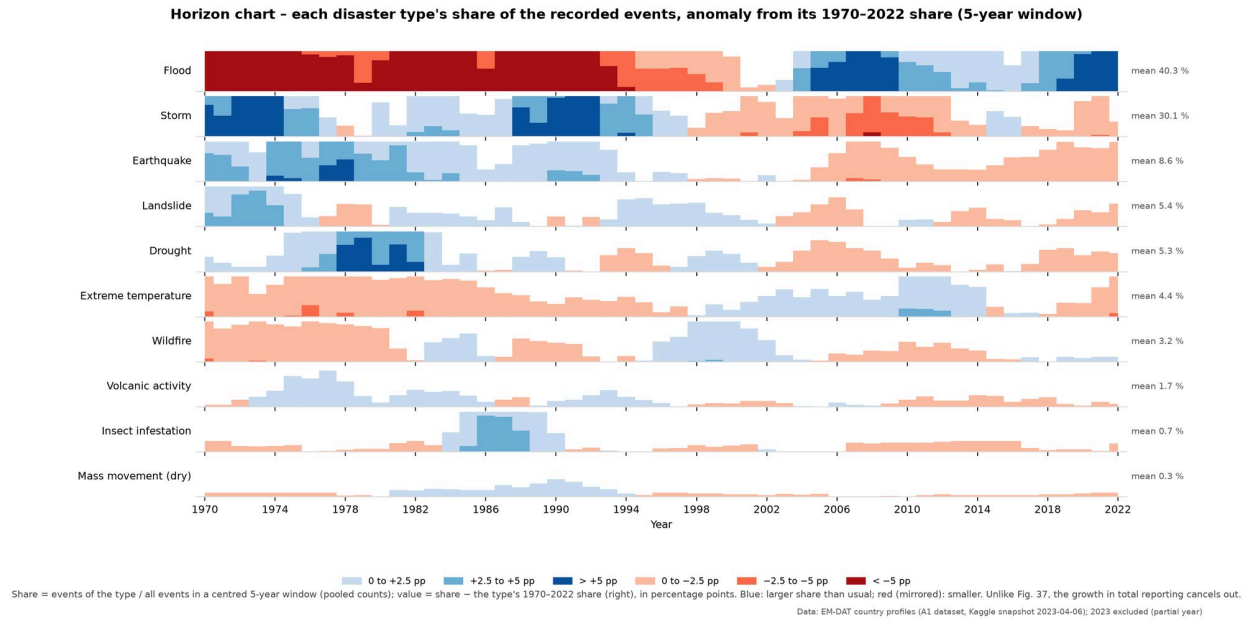


Fig. 38. Horizon chart of each disaster type's share of the recorded events (centred 5-year window) minus its 1970–2022 share, 1970–2022 (common scale, bands of 2.5 pp; blue above, red mirrored below; mean share on the right).

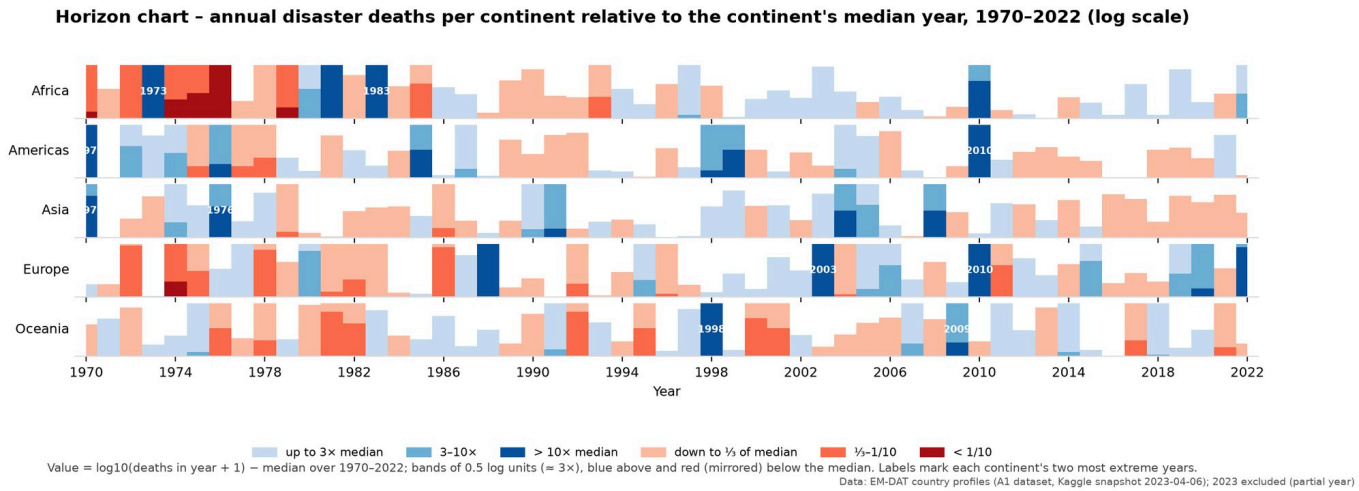


Fig. 39. Horizon chart of annual deaths per continent relative to the continent's median year (log scale; blue above, red mirrored below), 1970–2022.

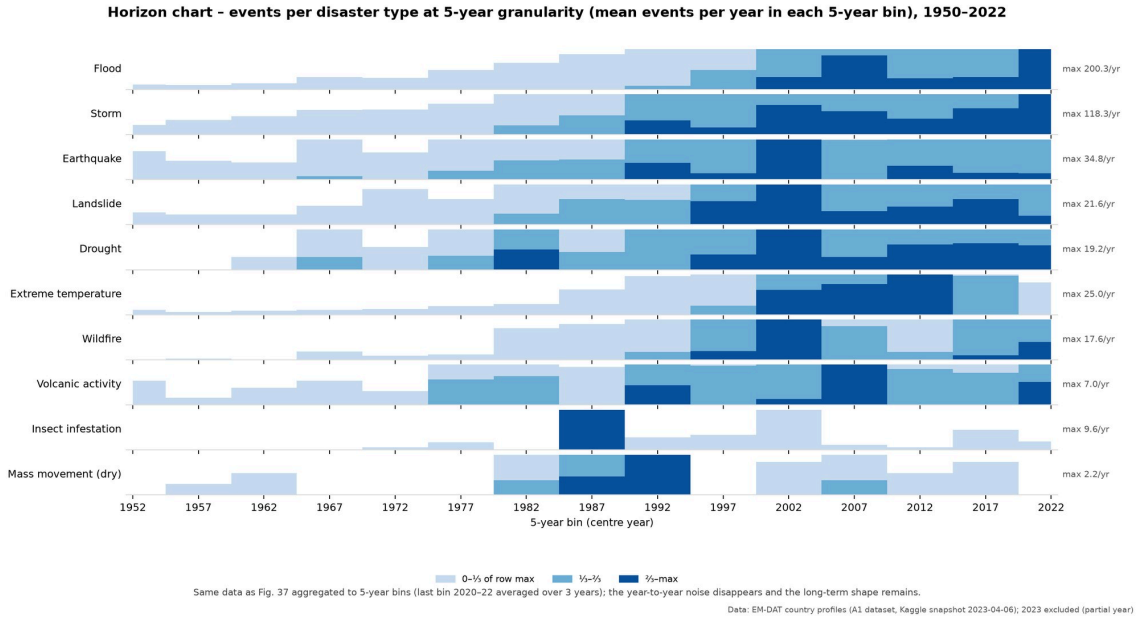


Fig. 40. Horizon chart of events per disaster type at 5-year granularity (mean events per year in each bin), 1950–2022.

C. Ridgeline plots

Fig. 41 shows, for each type, the distribution of the years in which its events occurred, 1900–2022. Sorted by median year, the types form a clear sequence: insect infestations (1988), dry mass movements (1990), earthquakes (1993), volcanic activity (1996), droughts (1999), landslides (2000), storms (2001), wildfires (2002), extreme temperature (2005) and floods (2006). Floods and temperature extremes are the "youngest" part of the record. Earthquakes and volcanoes have long tails back to 1900, and insect infestations form a sharp pulse in the late 1980s. Fig. 42 switches to decades as groups. The distribution of deaths per event moves left by almost an order of magnitude, with medians of 92 (1950s), 59, 38, 33, 18.6, 12.8 and 12.7 (2010s). This reflects real improvements in warning and preparedness, but also the registration of many more small events: the share of records with zero or unreported deaths rises from 8.5 % in the 1950s to about 30 % from the 1970s onwards. Fig. 43 uses continents as groups. The median number of people affected per event is 1,200 in Europe, 2,500 in Oceania, 5,000 in the Americas, 20,600 in Africa and 41,900 in Asia. The 35-fold gap between Europe and Asia reflects population density, exposure and vulnerability, and Africa's wide distribution reflects droughts that affect millions of people at once.

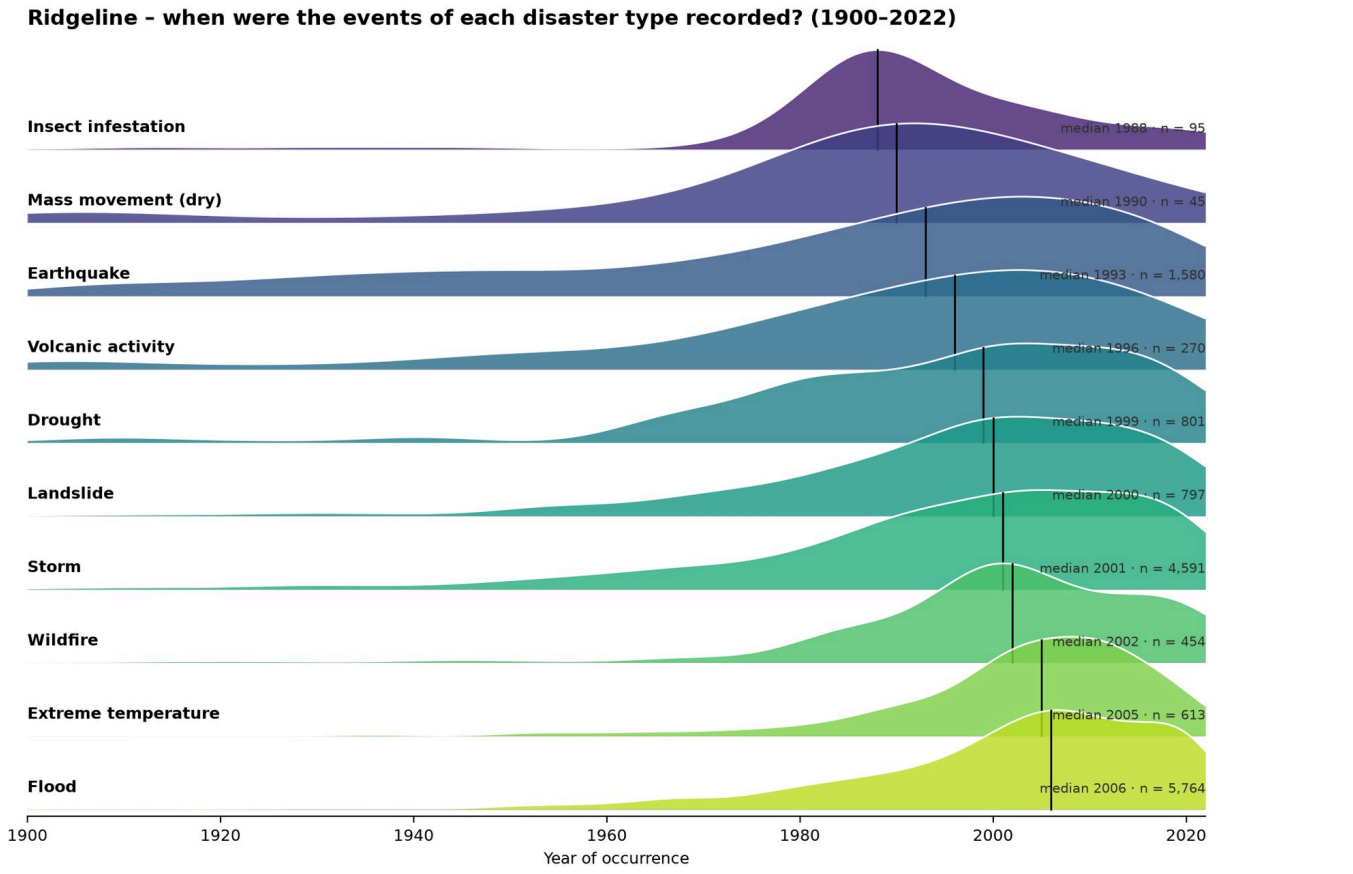


Fig. 41. Ridgeline plot of the year of occurrence of the events of each disaster type (1900–2022, weighted by events; black tick = median).

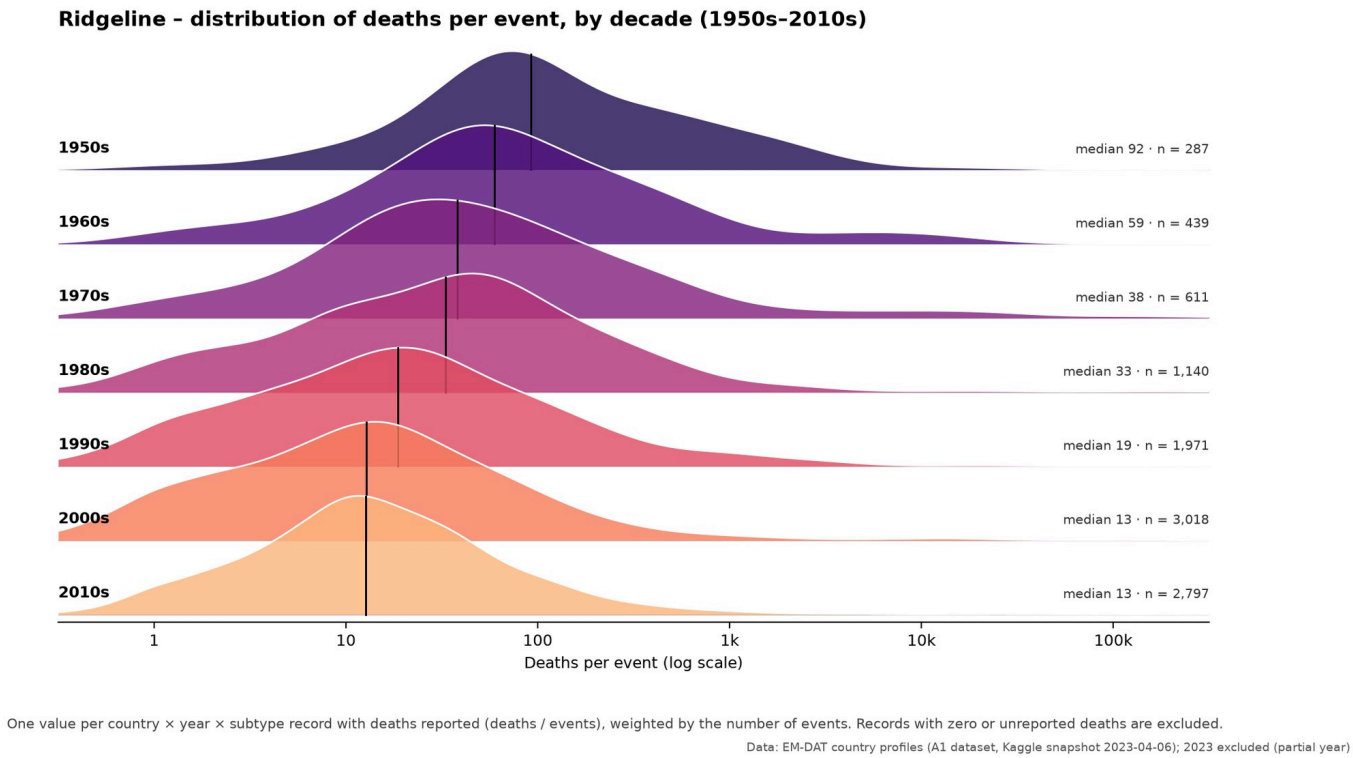


Fig. 42. Ridgeline plot of deaths per event by decade, 1950s–2010s (log scale).

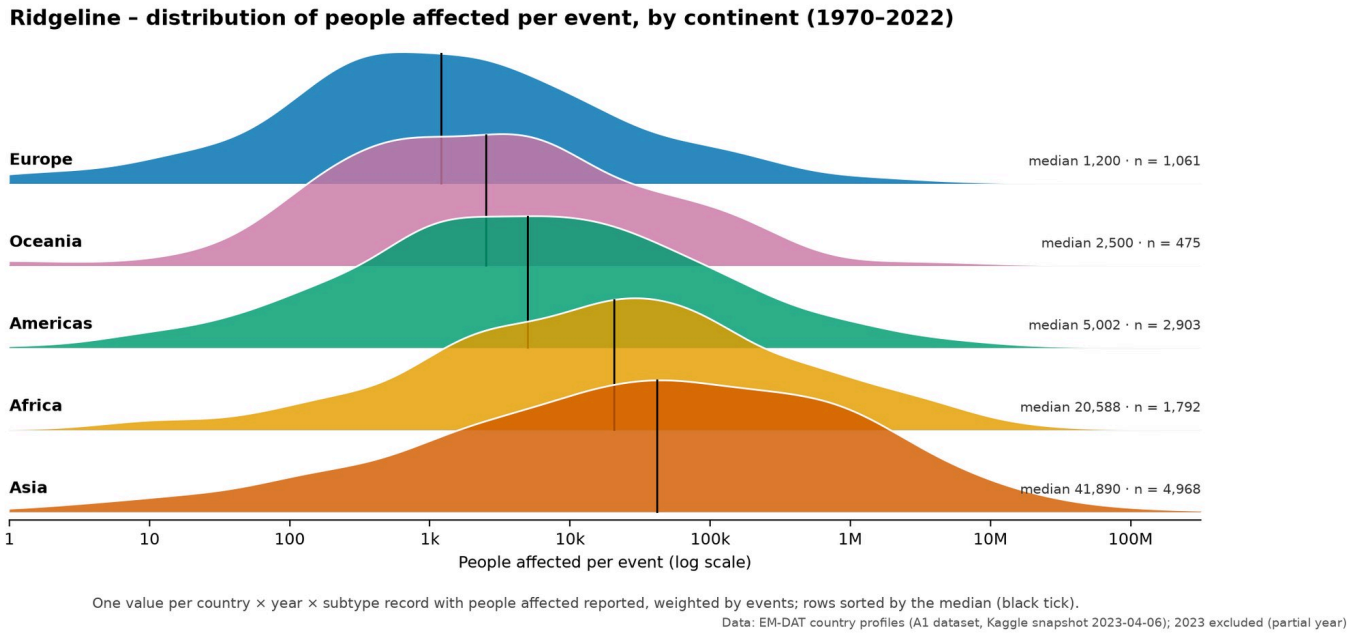


Fig. 43. Ridgeline plot of people affected per event by continent, 1970–2022 (log scale).

IX. Discussion

Parameter choices decide what is seen. In every task one parameter changed the message: global vs. local normalisation and linear vs. log scaling (Sec. III), the isovalue tie rule and smoothing (Sec. IV), sampling stride, arrow scaling and seeding (Sec. V), k and the similarity threshold (Sec. VI), bundling strength and the treemap measure (Sec. VII), and per-row vs. common scaling (Sec. VIII). We therefore state these parameters in each caption and keep them identical across compared panels.

Visualization exposes data artefacts. Several findings concern the data, not the phenomena: quantisation ties in VPD, storage gaps in precipitation, the underestimated storm maxima of a 16 km subsample, damage-reporting bias, state-succession "growth", the partial year 2023 and the geophysical bias of early records. Tools can introduce artefacts too: vectors given in longitude–latitude must be converted before cartopy projects them (Sec. II-C). Handling them (Tables III–IV) is a precondition for correct inferences.

Interaction is essential for InfoVis. Brushing turned the PCP into a query tool (Fig. 30). Zooming in the treemap revealed that a few famines dominate the century-long death toll (Fig. 33). Filtering and focus reduced the network to readable backbones (Fig. 27).

Limitations. The reconstructed daily-mean wind is an approximation and, over land only, cannot show a hurricane's closed circulation (Fig. 22). Ten uniform samples miss short extremes such as Barry and Dorian, which we show separately. The country-profile export has no event-level records, months or locations, which limits temporal granularity to years. The network depends on k and the BC threshold, and Louvain is stochastic (we fix the seed). Bundled PCPs distort values between axes. Finally, our quality measures are computational, not results of a user study.

X. Conclusion

The gridMET slice for July–September 2019 showed the seasonal retreat of heat and dry air: the area with VPD ≥ 3 kPa fell from 0.76 M km² to zero. It also captured a heat wave, Tropical Storm Imelda and an early cold outbreak; 4 km close-ups added the storms between the samples and showed that Imelda's record rain came from weak but convergent winds. The tasks showed that colour-map normalisation, contour thresholds and vector sampling must be chosen deliberately to show such features faithfully. The EM-DAT analyses showed that disaster regimes follow climate and tectonics rather than continents; that frequency, mortality, people affected and damage rank hazards differently; and that a large part of the apparent growth in disasters is a property of the record itself, while the shares of floods and temperature extremes did grow. Each visualization is reproducible from the scripts in `code/` (see the README).

Contributions of Team Members

Lakshya Jain (BT2024044) designed and implemented SciVis Task 1, colour mapping (`scivis/color_mapping.py`; Figs. 2–9 and four GIFs), and InfoVis Task 2: the multivariate and hierarchical data builder (`infovis/build_web_data.py`), the JavaScript parallel-coordinates plot with edge bundling and the zoomable treemap (`web/parallel_coordinates.html`, `web/treemap.html`, `js/pcp.js`, `js/treemap.js`; Figs. 28–36). He wrote Sections III and VII.

Kalpit (BT2024093) designed and implemented SciVis Task 2, contour mapping: the Marching Squares implementation and its validation, and the contour-fill, smoothing and temporal experiments (`scivis/marching_squares.py`, `scivis/contour_mapping.py`; Figs. 10–15 and a GIF). He also built InfoVis Task 1: network construction, the NetworkX layouts and metrics, the collision-free labelling, the Gephi export and the interactive viewer

(`infovis/network_layouts.py`, `web/network.html`, `js/network.js`; Figs. 23–27). He wrote Sections IV and VI.

Arnav (BT2024233) designed and implemented SciVis Task 3: quiver plots, the grid-sampling and arrow-scaling experiments, the latitude correction of the vector transform, streamlines with four seeding strategies and the storm close-ups (`scivis/quiver_streamlines.py`; Figs. 16–22 and two GIFs). He also built InfoVis Task 3, the horizon and ridgeline plots (`infovis/horizon_ridgeline.py`; Figs. 37–43). He wrote Sections V and VIII.

Joint work. Data download, cleaning and processing for both datasets, and the shared utilities (`data_prep/*`, `scivis/common.py`, `web/js/common.js`, `tools/*`; Fig. 1), were done jointly. So were Sections I, II, IX and X and the README.

References

[1] J. T. Abatzoglou, "Development of gridded surface meteorological data for ecological applications and modelling," *Int. J. Climatology*, vol. 33, no. 1, pp. 121–131, 2013.

[2] Climatology Lab, "gridMET," https://www.climatologylab.org/gridmet.html (accessed Oct. 2026).

[3] D. Guha-Sapir, R. Below, and Ph. Hoyois, "EM-DAT: The CRED/OFDA International Disaster Database," Université catholique de Louvain, Brussels, Belgium, www.emdat.be.

[4] mexwell, "Natural Disasters (Emergency Events Database)," Kaggle, https://www.kaggle.com/datasets/mexwell/natural-disasters-emergency-events-database (accessed Oct. 2026).

[5] S. Hoyer and J. Hamman, "xarray: N-D labeled arrays and datasets in Python," *J. Open Research Software*, vol. 5, no. 1, p. 10, 2017.

[6] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[7] Met Office, "Cartopy: a cartographic Python library with a Matplotlib interface," https://scitools.org.uk/cartopy.

[8] National Hurricane Center, "Tropical Cyclone Reports – 2019 Atlantic Hurricane Season," NOAA, https://www.nhc.noaa.gov/data/tcr/.

[9] D. Borland and R. M. Taylor II, "Rainbow color map (still) considered harmful," *IEEE Computer Graphics and Applications*, vol. 27, no. 2, pp. 14–17, 2007.

[10] K. Moreland, "Diverging color maps for scientific visualization," in *Advances in Visual Computing (ISVC)*, LNCS 5876, 2009, pp. 92–103.

[11] M. Harrower and C. A. Brewer, "ColorBrewer.org: An online tool for selecting colour schemes for maps," *The Cartographic Journal*, vol. 40, no. 1, pp. 27–37, 2003.

[12] P. Kovesi, "Good colour maps: How to design them," arXiv:1509.03700, 2015.

[13] S. van der Walt and N. Smith, "A better default colormap for Matplotlib," talk at SciPy 2015, Austin, TX, 2015.

[14] W. E. Lorensen and H. E. Cline, "Marching cubes: A high resolution 3D surface construction algorithm," *ACM SIGGRAPH Computer Graphics*, vol. 21, no. 4, pp. 163–169, 1987.

[15] ContourPy developers, "ContourPy: Python library for calculating contours of 2D quadrilateral grids," https://contourpy.readthedocs.io.

[16] B. Jobard and W. Lefer, "Creating evenly-spaced streamlines of arbitrary density," in *Visualization in Scientific Computing '97*, Springer, 1997, pp. 43–55.

[17] G. Turk and D. Banks, "Image-guided streamline placement," in *Proc. SIGGRAPH '96*, 1996, pp. 453–460.

[18] D. H. Laidlaw *et al.*, "Comparing 2D vector field visualization methods: A user study," *IEEE Trans. Vis. Comput. Graphics*, vol. 11, no. 1, pp. 59–70, 2005.

[19] K. E. Mitchell *et al.*, "The multi-institution North American Land Data Assimilation System (NLDAS)," *J. Geophys. Res.*, vol. 109, D07S90, 2004.

[20] A. Bhattacharyya, "On a measure of divergence between two statistical populations defined by their probability distributions," *Bull. Calcutta Math. Soc.*, vol. 35, pp. 99–109, 1943.

[21] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *J. Stat. Mech.*, P10008, 2008.

[22] T. M. J. Fruchterman and E. M. Reingold, "Graph drawing by force-directed placement," *Software: Practice and Experience*, vol. 21, no. 11, pp. 1129–1164, 1991.

[23] T. Kamada and S. Kawai, "An algorithm for drawing general undirected graphs," *Information Processing Letters*, vol. 31, no. 1, pp. 7–15, 1989.

[24] M. Jacomy, T. Venturini, S. Heymann, and M. Bastian, "ForceAtlas2, a continuous graph layout algorithm for handy network visualization designed for the Gephi software," *PLoS ONE*, vol. 9, no. 6, e98679, 2014.

[25] Y. Koren, "Drawing graphs by eigenvectors: theory and practice," *Computers & Mathematics with Applications*, vol. 49, no. 11–12, pp. 1867–1888, 2005.

[26] A. A. Hagberg, D. A. Schult, and P. J. Swart, "Exploring network structure, dynamics, and function using NetworkX," in *Proc. 7th Python in Science Conf.*, 2008, pp. 11–15.

[27] M. Bostock, V. Ogievetsky, and J. Heer, "D³: Data-driven documents," *IEEE Trans. Vis. Comput. Graphics*, vol. 17, no. 12, pp. 2301–2309, 2011.

[28] A. Inselberg, "The plane with parallel coordinates," *The Visual Computer*, vol. 1, no. 2, pp. 69–91, 1985.

[29] K. T. McDonnell and K. Mueller, "Illustrative parallel coordinates," *Computer Graphics Forum*, vol. 27, no. 3, pp. 1031–1038, 2008.

[30] G. Palmas, M. Bachynskyi, A. Oulasvirta, H.-P. Seidel, and T. Weinkauf, "An edge-bundling layout for interactive parallel coordinates," in *Proc. IEEE PacificVis*, 2014, pp. 57–64.

[31] J. Heinrich and D. Weiskopf, "State of the art of parallel coordinates," in *Eurographics 2013 – State of the Art Reports*, 2013, pp. 95–116.

[32] B. Johnson and B. Shneiderman, "Tree-maps: A space-filling approach to the visualization of hierarchical information structures," in *Proc. IEEE Visualization '91*, 1991, pp. 284–291.

[33] M. Bruls, K. Huizing, and J. J. van Wijk, "Squarified treemaps," in *Data Visualization 2000*, Springer, 2000, pp. 33–42.

[34] T. Saito, H. N. Miyamura, M. Yamamoto, H. Saito, Y. Hoshiya, and T. Kaseda, "Two-tone pseudo coloring: Compact visualization for one-dimensional data," in *Proc. IEEE InfoVis*, 2005, pp. 173–180.

[35] J. Heer, N. Kong, and M. Agrawala, "Sizing the horizon: The effects of chart size and layering on the graphical perception of time series visualizations," in *Proc. ACM CHI*, 2009, pp. 1303–1312.

[36] C. O. Wilke, *Fundamentals of Data Visualization*. O'Reilly Media, 2019.

[37] M. Gall, K. A. Borden, and S. L. Cutter, "When do losses count? Six fallacies of natural hazards loss data," *Bull. Amer. Meteor. Soc.*, vol. 90, no. 6, pp. 799–809, 2009.

[38] D. W. Scott, *Multivariate Density Estimation: Theory, Practice, and Visualization*. Wiley, 1992.

[39] M. Bastian, S. Heymann, and M. Jacomy, "Gephi: An open source software for exploring and manipulating networks," in *Proc. Int. AAAI Conf. Weblogs and Social Media (ICWSM)*, 2009, pp. 361–362.